Semiconductor Electronics: A Lecture Course

ECE 340, Spring 2026, ZJUI Institute (Prof. Said Mikki)

Contents

1 Introduction - Why Semiconductors Matter **3**
1.1 Course Philosophy: From First Principles to Device Operation . . . 3
1.2 The Central Goal: Mastering the Drift-Diffusion-Poisson Model . . . 4
1.3 The Drift-Diffusion Model as a Unifying Framework . . . 4
1.4 Prerequisites and Mathematical Background . . . 5
1.5 The Engine of the Information Age . . . 5
1.6 Beyond Computing: A Plethora of Applications . . . 6
1.7 The Interdisciplinary Nature of Semiconductor Engineering . . . 7

2 Crystal Structures: The Foundation of Semiconductor Devices **9**
2.1 What is a Crystal? . . . 9
2.2 Atomic Bonding in Semiconductors . . . 9
2.3 Essential Quantum Mechanics Review . . . 9

3 Energy Bands: From Discrete Atoms to Crystals **11**
3.1 Electron Motion in a Periodic Potential . . . 11
3.2 Energy Splitting: The Resonance Analogy . . . 11
3.3 From Discrete Levels to Energy Bands . . . 12
3.4 The Bandgap and Semiconductor Classification . . . 13
3.5 Why Semiconductors are Special . . . 14

4 Dispersion Relations: Connecting Energy and Momentum **15**
4.1 The Free Electron Dispersion Relation . . . 15
4.2 Electrons in a Periodic Potential: The Effective Mass Approximation . . . 16
4.3 Effective Mass: Curvature is Mass . . . 17
4.4 Where Do Dispersion Relations Come From? . . . 18

5 Holes: The Missing Electrons? **19**
5.1 The Intuitive Picture: A Missing Electron . . . 19
5.2 Why Holes Matter in Semiconductors, Not in Metals . . . 19
5.3 Quantum Mechanical Foundation: Negative Effective Mass . . . 20
5.4 The Hole as a Positive Charge Carrier . . . 21
5.5 Alternative Justification of the Hole Concept: Current Argument . . . 22
5.6 Experimental Consequences . . . 23

6 Classification of Solids: Insulators, Conductors, and Semiconductors **24**
6.1 Band Structure Classification . . . 24
6.2 Detailed Analysis of Each Class . . . 24
6.3 Intrinsic Semiconductors . . . 25
6.4 Extrinsic Semiconductors and Doping . . . 26
6.5 Compensation and Carrier Statistics . . . 28

7 n-Type and p-Type Semiconductors **29**
7.1 n-Type Semiconductors . . . 29
7.2 p-Type Semiconductors . . . 30
7.3 Intuitive Comparison . . . 31

7.4 What About the Fermi Level? . . . 31
7.5 Why This Matters . . . 31

8 The Fermi Energy: The "Fill Line" for Electrons 33
8.1 The Classical vs. Quantum Expectation . . . 33
8.2 Defining the Fermi Level . . . 34
8.3 Visualizing E_F Across Different Materials . . . 35
8.4 What Determines E_F and Why It Matters . . . 35
8.5 A Simple Rule of Thumb . . . 36

9 Density of States: Counting Available Quantum States 38
9.1 The Concept of Density of States . . . 38
9.2 Analogy: Parking Spaces in a Parking Garage . . . 38
9.3 Density of States for Free Electrons . . . 39
9.4 Adapting the Formula for Semiconductors . . . 39
9.5 Valence Band Density of States and Holes . . . 40
9.6 Complete Picture: Conduction and Valence Bands Together . . . 41
9.7 Effective Density of States . . . 42
9.8 What's Next: The Occupation Probability . . . 42

10 Fermi-Dirac Distribution: The Probability of Occupation 43
10.1 Quantum Statistics: Fermions and Bosons . . . 43
10.2 The Three Fundamental Distribution Functions . . . 43
10.3 Mathematical Forms . . . 44
10.4 The Fermi-Dirac Distribution Function . . . 45
10.5 Key Properties . . . 45
10.6 Physical Interpretation . . . 46
10.7 The Maxwell-Boltzmann Approximation . . . 46
10.8 Summary: Computing Carrier Concentrations . . . 47

11 Equilibrium Carrier Concentrations 48
11.1 Electron Concentration in the Conduction Band . . . 48
11.2 Hole Concentration in the Valence Band . . . 50
11.3 Intuitive Picture: Where Do the Carriers Reside? . . . 52
11.4 The np Product and the Intrinsic Carrier Concentration . . . 55
11.5 Position of the Fermi Level . . . 56
11.6 Charge Neutrality and Exact Carrier Concentrations . . . 59
11.7 Practical Summary: When to Use Which Formula . . . 63
11.8 Comprehensive Summary of Key Equations . . . 64

1 Introduction - Why Semiconductors Matter

Welcome to ECE 340: Semiconductor Electronics. This course is built around a single unifying theme: the development of a rigorous, physically-based understanding of the **drift-diffusion model**—the fundamental set of equations that govern all semiconductor devices. Our journey will systematically derive the drift-diffusion-Poisson equations from first principles, then apply them to understand and analyze real devices.

Course Roadmap: Building the Drift-Diffusion Model (24 Lectures)

- **Lecture 1**: Introduction: Why semiconductors matter and course philosophy.
- **Lectures 2-3**: Band Theory Foundation: Dispersion relations $E(k)$, the concept of holes, effective mass m^*, and the distinction between conduction and valence bands.
- **Lectures 4-5**: Quantum Statistics and Density of States: Fermi-Dirac distribution, DoS, and the foundation for counting carriers.
- **Lectures 6-7**: Equilibrium Carrier Concentrations: Derivation of n and p in thermal equilibrium, the $np = n_i^2$ relation, and the position of the Fermi level.
- **Lecture 8**: Doping and Extrinsic Semiconductors: Donors, acceptors, n-type and p-type materials, and carrier concentrations in doped semiconductors.
- **Lectures 9-10**: Non-Equilibrium Distributions: Excess carriers, generation, recombination mechanisms, and the dynamics of charge relaxation.
- **Lecture 11**: Drift and Diffusion Currents: Derivation of drift current ($\mathbf{J} = qn\mu\mathbf{E}$), diffusion current ($\mathbf{J} = qD\nabla n$), and the Einstein relation $D/\mu = kT/q$.
- **Lectures 12-13**: The Drift-Diffusion-Poisson Model I: Continuity equations and current density equations — derivation and physical interpretation.
- **Lectures 14-15**: The Drift-Diffusion-Poisson Model II: Poisson's equation, coupling with continuity equations, and the complete semiconductor transport model.
- **Lectures 16-17**: p-n Junction in Equilibrium: Built-in potential, space charge region, and applying the drift-diffusion-Poisson model.
- **Lectures 18-19**: p-n Junction Under Bias: Derivation of the ideal diode equation $I = I_S(e^{qV/kT} - 1)$, junction capacitance, and breakdown.
- **Lectures 20-21**: Optical Devices: Photodiodes, solar cells, photoconductors, LEDs, and laser diodes — analyzed using the drift-diffusion framework with generation terms.
- **Lectures 22-23**: Bipolar Junction Transistors (BJTs): Structure, operating principles, current gain, Ebers-Moll model, and frequency response.
- **Lecture 24**: MOSFETs: Field-effect principle, threshold voltage, current-voltage characteristics, and body effect.

1.1 Course Philosophy: From First Principles to Device Operation

This course is structured around a deliberate, progressive derivation of the drift-diffusion-Poisson equations because this model represents the intellectual core of semiconductor device physics. Unlike a purely descriptive approach, our derivation-based methodology ensures that:

- Every equation we use has a clear physical origin—whether from quantum mechanics, statistical me-

chanics, or electromagnetism.

- The assumptions and approximations inherent in device models are explicitly identified and understood.
- Students gain the ability to derive results rather than merely memorize them.
- The unity of device physics becomes apparent: the same drift-diffusion equations, with appropriate boundary conditions, describe diodes, transistors, and optical devices.

1.2 The Central Goal: Mastering the Drift-Diffusion-Poisson Model

By the end of this course, you will have derived and understood the complete set of equations that form the foundation of all semiconductor device analysis:

$$\textbf{Current density equations:} \quad \mathbf{J}_n = q\mu_n n\mathbf{E} + qD_n\nabla n \tag{1}$$

$$\mathbf{J}_p = q\mu_p p\mathbf{E} - qD_p\nabla p \tag{2}$$

$$\textbf{Continuity equations:} \quad \frac{\partial n}{\partial t} = \frac{1}{q}\nabla\cdot\mathbf{J}_n + G - R \tag{3}$$

$$\frac{\partial p}{\partial t} = -\frac{1}{q}\nabla\cdot\mathbf{J}_p + G - R \tag{4}$$

$$\textbf{Poisson's equation:} \quad \nabla^2\phi = -\frac{\rho}{\epsilon} = -\frac{q}{\epsilon}(p - n + N_D^+ - N_A^-) \tag{5}$$

These five equations, properly understood and applied, constitute the complete description of semiconductor device operation. Every device we study—diodes, solar cells, BJTs, MOSFETs—represents a particular solution to this system under specific boundary conditions and operating regimes.

1.3 The Drift-Diffusion Model as a Unifying Framework

What makes the drift-diffusion-Poisson model so powerful is its universality. Whether analyzing a simple p-n junction or a complex MOSFET, the same physical principles apply:

- **Drift** describes carrier motion under electric fields—governed by mobility μ and effective mass m^*.
- **Diffusion** describes carrier motion from concentration gradients—governed by the diffusion coefficient D and the Einstein relation $D/\mu = kT/q$.
- **Generation and recombination** describe how carriers are created and destroyed—linking to optical absorption, thermal processes, and device lifetime.
- **Poisson's equation** couples the electrostatic potential to the charge distribution—closing the system and enabling self-consistent solutions.

The Unified Vision

By the conclusion of this course, you will recognize that:

- The **diode** is the simplest solution to the drift-diffusion equations in a p-n structure.
- The **BJT** represents coupled solutions in a three-terminal, two-junction structure.
- The **MOSFET** introduces field-effect control through the electrostatic boundary conditions.
- **Optical devices** add generation terms G to the continuity equations.

All emerge naturally from the same foundational equations—this is the elegance and power of the drift-diffusion model.

1.4 Prerequisites and Mathematical Background

To succeed in this course, you should be comfortable with:

- **Calculus**: Differentiation, integration, and basic differential equations.
- **Physics**: Basic electromagnetism (electric fields, potentials, Gauss's law).
- **Quantum mechanics** (optional but helpful): Basic concepts of energy levels and wave-particle duality will be reviewed as needed.
- **Statistical mechanics** (optional): The Fermi-Dirac distribution will be developed from first principles.

All necessary quantum and statistical concepts will be introduced as we build the drift-diffusion model from the ground up. No prior exposure to semiconductor physics is assumed.

What You Will Gain

Upon completing this course, you will be able to:

1. Derive the drift-diffusion-Poisson equations from fundamental physical principles.
2. Calculate equilibrium and non-equilibrium carrier concentrations in doped semiconductors.
3. Analyze p-n junctions, including the ideal diode equation, breakdown, and capacitance.
4. Design and analyze bipolar and field-effect transistors.
5. Understand the physics of optoelectronic devices including solar cells and photodetectors.
6. Apply the drift-diffusion framework to emerging device technologies.

1.5 The Engine of the Information Age

The device you are using to read this document—whether it's a laptop, a tablet, or a smartphone—is a testament to the triumph of semiconductor electronics. At the heart of every computer, server, and mobile device lies the integrated circuit (IC), a complex network of billions of tiny transistors fabricated on a single piece of semiconductor material, usually silicon.

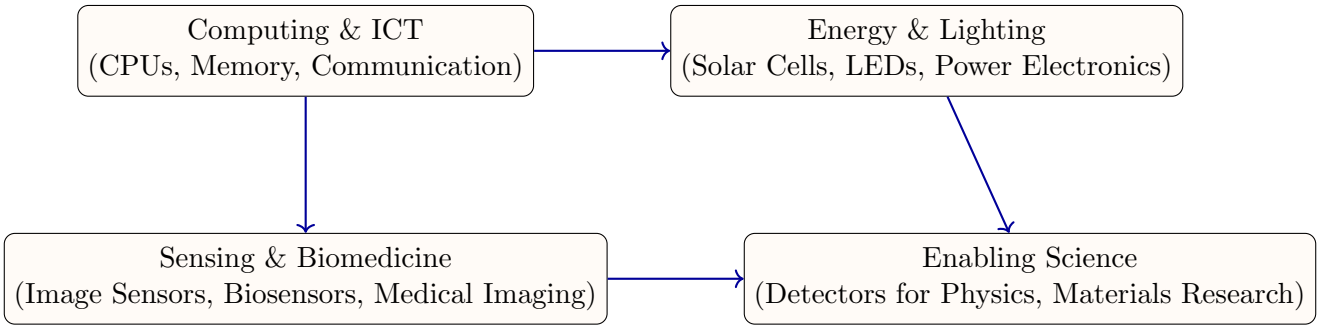
The semiconductor industry has relentlessly followed Moore's Law for decades, doubling the number of transistors on a chip approximately every two years. This has been achieved by continuously shrinking the size of transistors, which requires an ever-deeper understanding of quantum mechanics and materials science at the nanoscale—topics central to this course.

The Centrality of Semiconductors in Computing

- **The Physics of Computation**: All modern computing, from the simplest logic gate to the most complex AI processor, relies on the controlled flow of electrical charge through semiconductors. Understanding semiconductor physics means understanding the very **physical foundation of information processing**. You learn not just *how* to use a transistor, but *why* it works.
- **The Language of Integrated Circuits**: The entire ecosystem of microelectronics—including digital logic, memory (DRAM, SRAM, Flash), and processors (CPUs, GPUs)—is built upon semiconductor technology. A solid grasp of this field is essential for anyone aspiring to work in chip design, hardware engineering, or any related discipline.
- **The Unrivaled Technology**: For over half a century, semiconductor-based computing devices have been the only technology to achieve mass production and economic viability for general-purpose computing. While other approaches (like vacuum tubes, magnetic logic, or optical computing) have been explored, none have matured to challenge the scalability, cost-effectiveness, and energy efficiency of semiconductors. This course explains the unique material properties that make this dominance possible.

1.6 Beyond Computing: A Plethora of Applications

While computing is a giant of the field, the impact of semiconductors extends far beyond, enabling solutions in energy, health, and fundamental science.



- **Renewable Energy and Power Electronics**: The transition to a sustainable energy future is heavily reliant on semiconductors.
  - **Photovoltaic Solar Cells**: These devices are essentially large-area semiconductor diodes that convert light directly into electricity. The physics of the p-n junction, which we will study in detail, is the very principle that allows solar panels to generate power.
  - **Power Electronics**: Efficiently managing and converting electrical power from sources like solar panels and batteries requires power transistors (e.g., IGBTs, MOSFETs) made from semiconductors like silicon and, increasingly, wide-bandgap materials like silicon carbide (SiC) and gallium nitride (GaN).
- **Materials Science and Nanotechnology**: Semiconductors are a playground for materials science. By precisely engineering materials at the atomic scale—through techniques like doping, creating alloys (e.g., SiGe), and building nanostructures (quantum wells, wires, and dots)—we can design materials with entirely new electronic and optical properties. This course provides the foundational knowledge to understand these advanced material systems.

- **Sensing and the Internet of Things (IoT)**: Semiconductor sensors are everywhere. The CMOS image sensor in your phone's camera, the MEMS (Micro-Electro-Mechanical Systems) accelerometer that detects your screen's orientation, and chemical sensors for environmental monitoring are all semiconductor-based. Their small size, low cost, and low power consumption, all derived from semiconductor physics, make the IoT revolution possible.
- **Fundamental Physics and Scientific Discovery**: Semiconductor devices are indispensable tools in experimental physics. High-energy physics experiments, like those at CERN's Large Hadron Collider, use sophisticated silicon pixel detectors to track the paths of subatomic particles after collisions. Astrophysicists use semiconductor-based CCDs and infrared detectors in telescopes to study distant galaxies. Understanding these devices allows engineers to build better instruments for scientific discovery.

Why This Matters for You

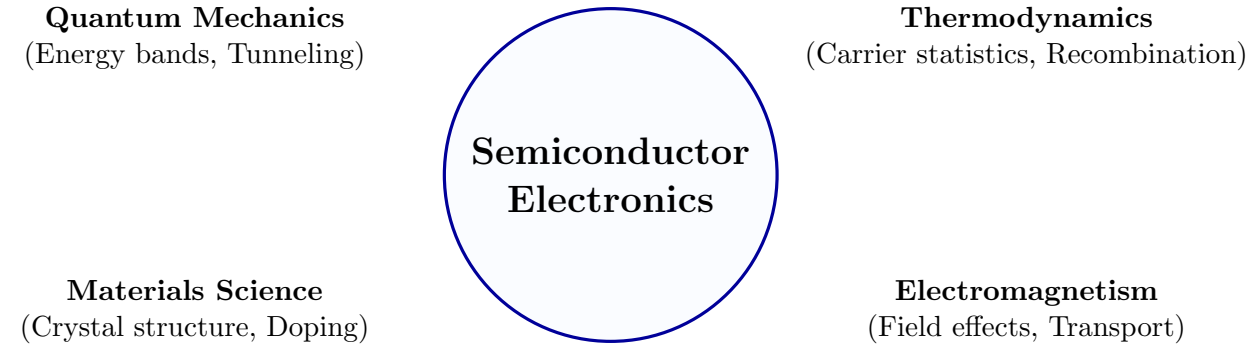
As an undergraduate in engineering or science, this course will equip you with a fundamental understanding of the materials that underpin your discipline.

- **For Electrical and Computer Engineers**: It is the core science of your profession. It connects the atomic-level properties of silicon to the macroscopic I-V curves of a transistor you will use in circuit design.
- **For Computer Scientists**: It demystifies the hardware. You will understand the physical "brick" upon which all algorithms and software are built, providing insight into the ultimate limits of computing performance.
- **For Mechanical Engineers**: It opens doors to the world of micro- and nano-systems (MEMS/NEMS), where mechanical structures are integrated with electronics, and to the growing field of smart materials and sensors.
- **For Physicists and Materials Scientists**: It is the premier applied example of quantum mechanics and solid-state physics, showing how fundamental principles lead to world-changing technologies.

This course will provide you with the essential toolkit to understand, innovate, and work with the technologies that define our modern world.

1.7 The Interdisciplinary Nature of Semiconductor Engineering

Success in semiconductor engineering requires a blend of physics, chemistry, and engineering principles. Throughout this course, we will draw on concepts from quantum mechanics, statistical thermodynamics, and electromagnetism to build a complete picture of how semiconductor devices operate.



This interdisciplinary nature is what makes the field so rich and impactful. As we proceed through the course, we will see how these different areas of knowledge converge to explain and enable the behavior of

real-world devices. The journey begins with understanding the fundamental quantum mechanics of electrons in solids, which is the focus of our next lecture.

2 Crystal Structures: The Foundation of Semiconductor Devices

Semiconductors are not amorphous materials like glass; they are **crystals**. To understand how a transistor or a diode works, we must first understand the ordered, periodic arrangement of atoms that gives semiconductors their unique electronic properties. This section provides the minimal essential knowledge of crystal structures and bonding as a foundation for the upcoming discussion on energy bands.

2.1 What is a Crystal?

Definition 1 (Crystal)**.** A crystal is a solid material whose constituents (atoms, molecules, or ions) are arranged in a highly ordered, repeating pattern that extends in all three spatial dimensions. This periodic arrangement is known as the **crystal lattice**.

Think of a crystal as a three-dimensional wallpaper pattern. Just as a wallpaper pattern is created by repeating a single tile, a crystal is built by repeating a fundamental unit called the **unit cell**.

- **Lattice**: A regular, periodic array of points in space.
- **Basis**: An atom or group of atoms attached to each lattice point.
- **Crystal Structure = Lattice + Basis**.

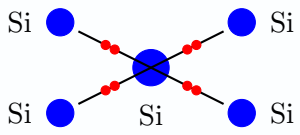
For semiconductors like silicon (Si) and germanium (Ge), the crystal structure is the **diamond cubic** structure. Compound semiconductors like gallium arsenide (GaAs) have a similar but slightly different structure called the **zincblende** structure. Both are characterized by each atom being covalently bonded to four nearest neighbors in a tetrahedral configuration.

2.2 Atomic Bonding in Semiconductors

The type of chemical bonding between atoms is crucial for determining whether a material is a semiconductor, a conductor, or an insulator.

Covalent Bonding in Silicon

Silicon, the workhorse of the semiconductor industry, has four valence electrons. In a silicon crystal, each atom shares its four valence electrons with its four nearest neighbors, forming **covalent bonds**. This sharing creates a stable configuration where each atom effectively has eight electrons in its outer shell (the "octet rule").



Shared valence electrons form covalent bonds

At absolute zero temperature ($T = 0$ K), all valence electrons are locked in these covalent bonds, and the material behaves as an insulator. As temperature increases, some electrons may gain enough thermal energy to break free from their bonds, becoming "free" to conduct electricity. This temperature-dependent conductivity is a hallmark of semiconductors.

2.3 Essential Quantum Mechanics Review

Before we can understand how electrons move through a crystal, we need a few fundamental concepts from quantum mechanics.

Definition 2 (Planck's Relation)**.** Energy and frequency are proportional. A photon (or any quantum particle) with frequency f has an energy E given by:

$$\boxed{E = hf = \hbar\omega}$$

where $h = 6.626 \times 10^{-34}$ J·s is Planck's constant, $\hbar = h/2\pi$, and $\omega = 2\pi f$ is the angular frequency.

Definition 3 (de Broglie Relation)**.** A particle with momentum p has an associated wavelength λ, called the de Broglie wavelength:

$$\boxed{\lambda = \frac{h}{p}} \quad \text{or} \quad \boxed{p = \hbar k}$$

where $k = 2\pi/\lambda$ is the wave number.

Definition 4 (Schrödinger Equation)**.** The behavior of a quantum particle (like an electron) is described by its wave function $\Psi(x, t)$, which satisfies the time-dependent Schrödinger equation:

$$\boxed{i\hbar\frac{\partial\Psi}{\partial t} = \hat{H}\Psi}$$

For a single particle in one dimension with potential energy $V(x)$, the Hamiltonian $\hat{H}$ is:

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)$$

The wave function itself is not directly measurable, but its square magnitude $|\Psi(x, t)|^2$ gives the probability of finding the particle at position x at time t.

Definition 5 (Electron Volt (eV))**.** The electron volt is a convenient unit of energy in semiconductor physics. It is defined as the kinetic energy gained by an electron accelerated through a potential difference of 1 volt:

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

Typical semiconductor bandgap energies are on the order of 1 eV (e.g., silicon: 1.12 eV, GaAs: 1.42 eV).

3 Energy Bands: From Discrete Atoms to Crystals

When atoms are brought together to form a crystal, their discrete energy levels evolve into continuous bands of allowed energies, separated by forbidden regions called **bandgaps**. This band structure determines whether a material is a conductor, semiconductor, or insulator.

3.1 Electron Motion in a Periodic Potential

A key question: How can an electron move freely through a densely packed crystal structure without constantly colliding with atoms? The answer lies in quantum mechanics.

Consider an electron entering a crystal. The atoms in the crystal create a periodic potential $V(x)$ that repeats with the lattice spacing a:

$$V(x+a) = V(x)$$

Bloch's Theorem

For an electron in a periodic potential, the Schrödinger equation has solutions of the form:

$$\boxed{\psi_k(x) = u_k(x)e^{ikx}}$$

where $u_k(x)$ is a function that has the same periodicity as the lattice: $u_k(x+a) = u_k(x)$. These are called **Bloch waves**.

The Bloch wave is a plane wave (e^{ikx}) modulated by a periodic function $u_k(x)$. This means:

- The electron is not localized; its wave function extends throughout the entire crystal.
- The probability density $|\psi_k(x)|^2 = |u_k(x)|^2$ is periodic, meaning the electron is equally likely to be found near any atom.
- Unlike a bound state that decays exponentially away from a nucleus (like in a hydrogen atom), the Bloch wave does **not decay**. It propagates through the crystal like a wave through a transmission line.

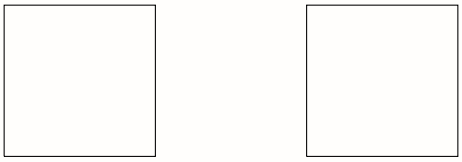
This is the quantum mechanical explanation for electron mobility in crystals: the periodic structure allows oscillatory, propagating solutions to the Schrödinger equation, not just decaying ones.

3.2 Energy Splitting: The Resonance Analogy

When atoms are far apart, they have discrete, sharp energy levels. As they come together to form a crystal, these levels split into bands. This phenomenon can be understood through a classical analogy.

Classical Analogy: Coupled Resonant Circuits

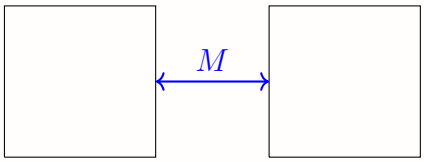
Consider two identical LC circuits, each with its own resonant frequency $f_0 = 1/(2\pi\sqrt{LC})$. When the circuits are isolated, they each oscillate at f_0.



Circuit 1: f_0 Circuit 2: f_0

Now, bring them close together so their inductors are coupled magnetically (mutual inductance M). The system now has two **normal modes** of oscillation:

- A **symmetric mode** where currents flow in the same direction: frequency $f_1 < f_0$
- An **antisymmetric mode** where currents flow in opposite directions: frequency $f_2 > f_0$



Result:
$f_1 < f_0$ (symmetric)
$f_2 > f_0$ (antisymmetric)

Key insight: The interaction (coupling) between two identical resonant systems causes the single resonant frequency to **split** into two distinct frequencies.

Now apply Planck's relation $E = hf$:

$$f_1 < f_0 < f_2 \quad \Rightarrow \quad E_1 < E_0 < E_2$$

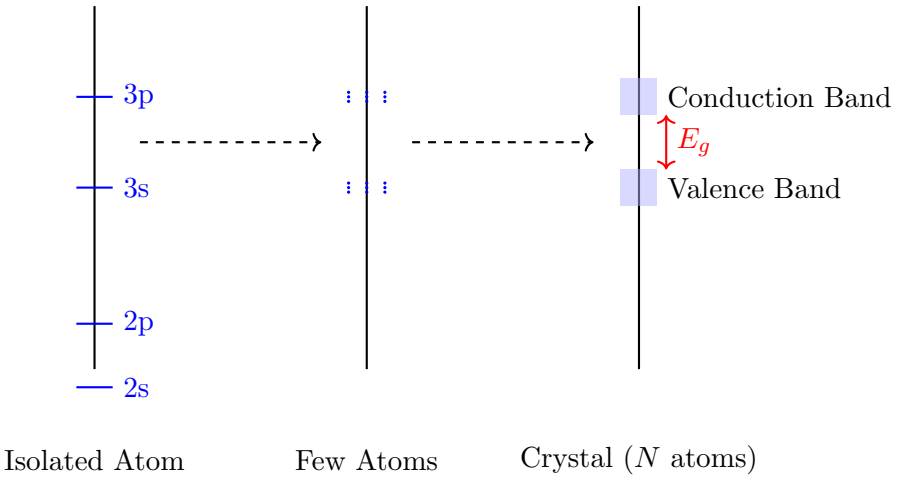
The single energy level E_0 of an isolated system splits into two distinct energy levels E_1 and E_2 when two identical systems are coupled.

3.3 From Discrete Levels to Energy Bands

Now generalize this to a crystal with N atoms:

- **Isolated atom**: Each electron orbital has a discrete energy level (e.g., 1s, 2s, 2p).
- **Two atoms**: When two atoms bond, each energy level splits into two closely spaced levels (one lower, one higher), corresponding to bonding and antibonding states.
- **Many atoms (N very large)**: When N atoms come together to form a crystal, each original atomic level splits into N discrete but very closely spaced energy levels.

Since N is enormous (typically $\sim 10^{22}$ atoms/cm^3), these N levels are so close together that they form a quasi-continuous **energy band**.



3.4 The Bandgap and Semiconductor Classification

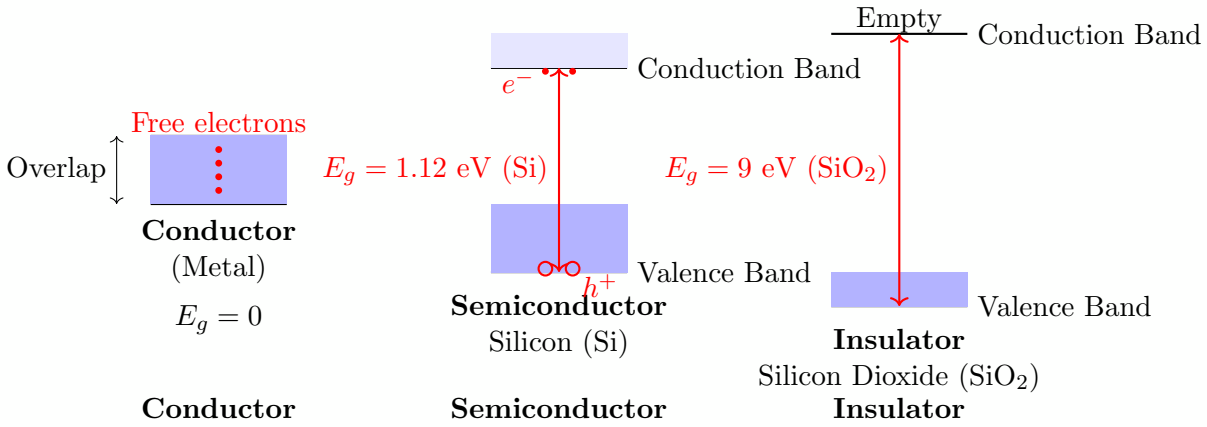
The splitting creates two important bands:

- **Valence Band**: The lower band, formed from the splitting of the outermost (valence) electron orbitals. At low temperatures, this band is fully occupied by electrons.
- **Conduction Band**: The upper band, formed from the splitting of the next-higher energy orbitals (the ones that would be empty in isolated atoms). This band is empty at low temperatures.
- **Bandgap (E_g)**: The forbidden energy region between the top of the valence band and the bottom of the conduction band. No electron states exist here.

Bandgap Energies for Common Semiconductors

Material	Symbol	Bandgap E_g at 300 K (eV)
Germanium	Ge	0.67
Silicon	Si	1.12
Gallium Arsenide	GaAs	1.42
Gallium Nitride	GaN	3.4
Silicon Carbide	SiC	3.3

Visual representation of the energy band diagram:



Note: Bandgap values are material-dependent
Si: 1.12 eV, Ge: 0.67 eV, GaAs: 1.42 eV, SiO_2: ∼9 eV

3.5 Why Semiconductors are Special

The key to semiconductor behavior lies in the bandgap size:

- **Conductors (Metals)**: The valence and conduction bands overlap ($E_g = 0$). Electrons can move freely without needing extra energy.
- **Insulators**: The bandgap is very large ($E_g > 3-4$ eV). At room temperature, essentially no electrons have enough thermal energy to jump from the valence band to the conduction band.
- **Semiconductors**: The bandgap is intermediate (~ 1 eV). At room temperature, a small but significant number of electrons gain enough thermal energy ($k_BT \approx 0.026$ eV at 300 K) to be excited across the gap, leaving behind "holes" in the valence band and enabling conduction.

This ability to control conductivity through temperature, doping, and electric fields is what makes semiconductors the foundation of modern electronics. In the next sections, we will explore how we can engineer these properties to create useful devices.

4 Dispersion Relations: Connecting Energy and Momentum

The single most important input from quantum theory to semiconductor physics is the **dispersion relation**. This mathematical relationship between the energy E and momentum p (or wave number k) of an electron determines virtually all of its dynamic properties—how it moves, how it responds to electric fields, and how it interacts with its environment.

4.1 The Free Electron Dispersion Relation

Let us begin with the simplest possible case: a completely free electron in empty space, with no forces acting on it.

Free Electron Schrödinger Equation

For a free electron, the potential energy $V(x) = 0$. The Hamiltonian consists only of kinetic energy:

$$\hat{H} = \frac{\hat{p}^2}{2m_e} = -\frac{\hbar^2}{2m_e}\frac{d^2}{dx^2}$$

The time-independent Schrödinger equation is:

$$-\frac{\hbar^2}{2m_e}\frac{d^2\psi}{dx^2} = E\psi$$

The solution to this equation is the plane wave:

$$\boxed{\psi(x) = Ae^{ikx} + Be^{-ikx}}$$

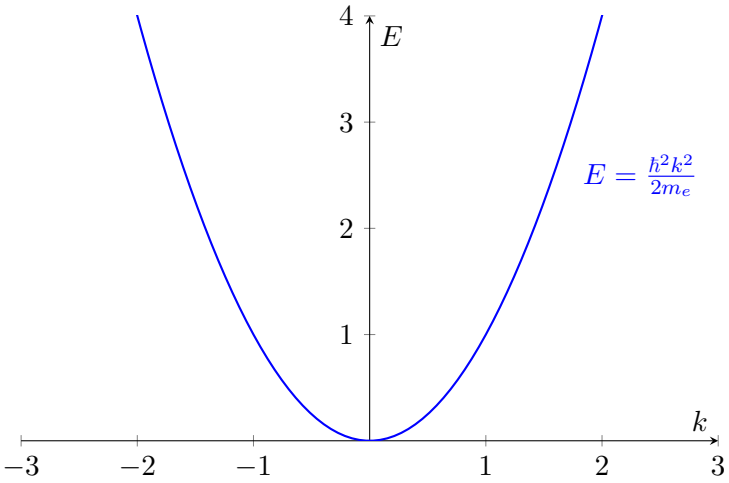
Recall from de Broglie's relation that the wave number k is related to momentum by $p = \hbar k$. Substituting the plane wave solution $\psi(x) = e^{ikx}$ into the Schrödinger equation:

$$-\frac{\hbar^2}{2m_e}\frac{d^2}{dx^2}e^{ikx} = -\frac{\hbar^2}{2m_e}(ik)^2e^{ikx} = \frac{\hbar^2k^2}{2m_e}e^{ikx} = Ee^{ikx}$$

This yields the **free electron dispersion relation**:

$$\boxed{E = \frac{\hbar^2k^2}{2m_e}}$$

Free Electron Parabolic E-k Relation



This parabolic relationship is fundamental. It tells us that:

- Energy increases quadratically with wave number (momentum).
- For a given $|k|$, there are two states: one with positive momentum ($+k$) and one with negative momentum ($-k$).
- The curvature of the parabola is determined by the electron mass m_e.

While this derivation assumed a completely free electron, and electrons in solids are clearly *not* free, this parabolic relation serves as the conceptual foundation for all that follows. The modifications introduced by the crystal lattice will preserve this basic quadratic form but with crucial adjustments.

4.2 Electrons in a Periodic Potential: The Effective Mass Approximation

In a real crystal, the electron moves through a complex potential created by the positively charged atomic nuclei and all the other electrons. This potential $V(\mathbf{r})$ is periodic with the lattice spacing:

$$V(\mathbf{r}+\mathbf{R}) = V(\mathbf{r})$$

where $\mathbf{R}$ is any lattice translation vector. The full Schrödinger equation becomes extremely complicated—far too complex to solve exactly for a real crystal with $\sim 10^{23}$ interacting particles.

The Independent Electron Approximation

To make progress, we use the **independent electron approximation**. The key idea is this: we assume that each electron propagates independently through the crystal, but with all the effects of the other electrons and the nuclei absorbed into an **effective background potential** $V_{\text{eff}}(\mathbf{r})$.

$$\left[-\frac{\hbar^2}{2m_e}\nabla^2 + V_{\text{eff}}(\mathbf{r})\right]\psi(\mathbf{r}) = E\psi(\mathbf{r})$$

This effective potential is still periodic, but it incorporates:

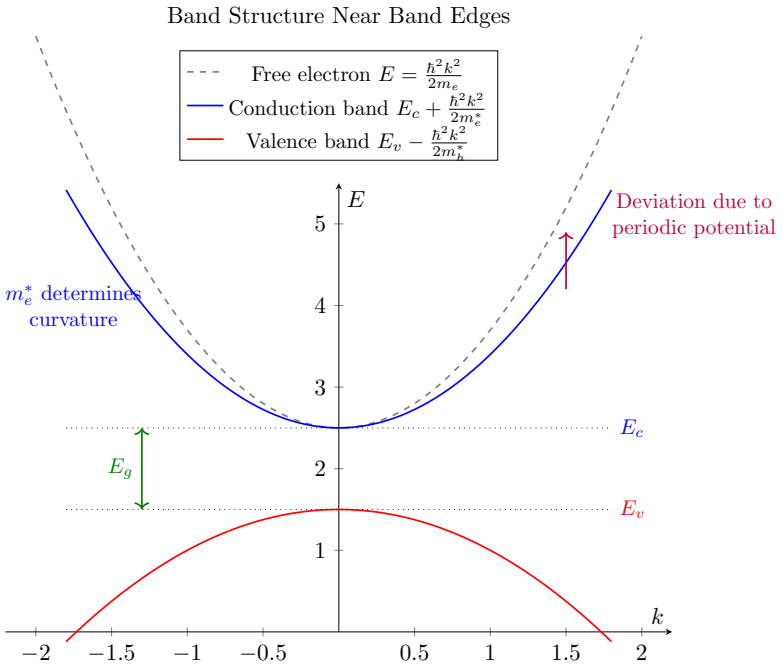
- The average repulsive interaction with all other electrons
- The attractive potential from the nuclei
- Exchange and correlation effects (quantum mechanical interactions between electrons)

Solving this equation (using methods from solid state physics like the nearly-free electron model or tight-binding model) yields the actual band structure $E(k)$ for the crystal. Remarkably, near the band edges (the bottom of the conduction band or the top of the valence band), this $E(k)$ relationship can often be approximated as parabolic, but with a modified curvature.

$$\boxed{E(k) \approx E_c + \frac{\hbar^2 k^2}{2m_e^*}} \quad \text{(near conduction band minimum)}$$

$$\boxed{E(k) \approx E_v - \frac{\hbar^2 k^2}{2m_h^*}} \quad \text{(near valence band maximum)}$$

Here, m_e^* and m_h^* are the **effective masses** for electrons and holes (missing electrons in the valence band), respectively. E_c is the conduction band edge energy, and E_v is the valence band edge energy.



4.3 Effective Mass: Curvature is Mass

The effective mass is not just a fitting parameter—it has a rigorous and profound definition rooted in the band structure itself.

Effective Mass Definition

Start with the free electron dispersion relation:

$$E = \frac{\hbar^2 k^2}{2m_e}$$

Differentiate once with respect to k:

$$\frac{dE}{dk} = \frac{\hbar^2 k}{m_e} = \frac{\hbar p}{m_e} = \hbar v_g$$

This gives the **group velocity** $v_g = (1/\hbar)(dE/dk)$.
Differentiate a second time:

$$\frac{d^2E}{dk^2} = \frac{\hbar^2}{m_e}$$

Solving for mass:

$$\boxed{m_e = \frac{\hbar^2}{d^2E/dk^2}}$$

For electrons in a crystal, we *define* the effective mass by this same relationship:

$$\boxed{m^* = \frac{\hbar^2}{d^2E/dk^2}}$$

The effective mass is therefore a measure of the **curvature** of the $E(k)$ dispersion relation:

- **Large curvature** (d^2E/dk^2 large) $\Rightarrow$ **small effective mass** — electron responds easily to forces

- **Small curvature** (d^2E/dk^2 small) $\Rightarrow$ **large effective mass** — electron is "heavy" and sluggish
- **Negative curvature** ($d^2E/dk^2 < 0$) $\Rightarrow$ **negative effective mass** — occurs near the top of the valence band and gives rise to holes (positively charged quasi-particles)

Example 1 (Electron and Hole Effective Masses in Common Semiconductors)**.**

Material	**Electron effective mass** (m_e^*/m_0)	**Hole effective mass** (m_h^*/m_0)
Silicon (Si)	0.98 (longitudinal) 0.19 (transverse)	0.49 (heavy) 0.16 (light)
Germanium (Ge)	1.64 (longitudinal) 0.082 (transverse)	0.28 (heavy) 0.044 (light)
Gallium Arsenide (GaAs)	0.067	0.45 (heavy) 0.082 (light)

Note: $m_0 = 9.11 \times 10^{-31}$ kg is the free electron rest mass.

4.4 Where Do Dispersion Relations Come From?

In practice, the detailed $E(k)$ dispersion relations for real materials come from two sources:

1. **Ab initio numerical calculations**: Using powerful computers and density functional theory (DFT) or other first-principles methods, solid state physicists can solve the Schrödinger equation for the periodic potential to high accuracy. These calculations yield the full band structure $E(k)$ for all k in the Brillouin zone.
2. **Experimental measurement**: Techniques like angle-resolved photoemission spectroscopy (ARPES) can directly measure the $E(k)$ relationship by shining X-rays on a material and analyzing the energy and angle of emitted electrons.

From these dispersion relations, we estimate effective masses by numerically computing the second derivative d^2E/dk^2 near the band edges. These effective masses then become essential parameters for device modeling, allowing us to treat electrons in semiconductors using modified versions of classical equations, but with m_e replaced by m^*.

Why Effective Mass Matters: A glimpse into future topics

The effective mass appears in virtually every important equation in semiconductor device physics:

- **Density of states**: $g_c(E) = \frac{(2m_e^*)^{3/2}}{2\pi^2\hbar^3}\sqrt{E - E_c}$
- **Mobility**: $\mu = \frac{q\tau}{m^*}$ (where τ is the scattering time)
- **Drift velocity**: $\mathbf{v}_d = \mu\mathbf{E} = \frac{q\tau}{m^*}\mathbf{E}$
- **Thermal velocity**: $v_{th} = \sqrt{\frac{3k_BT}{m^*}}$

A smaller effective mass means higher mobility, faster devices, but also lighter, more quantum-mechanically delocalized electrons. This trade-off is central to semiconductor materials engineering.

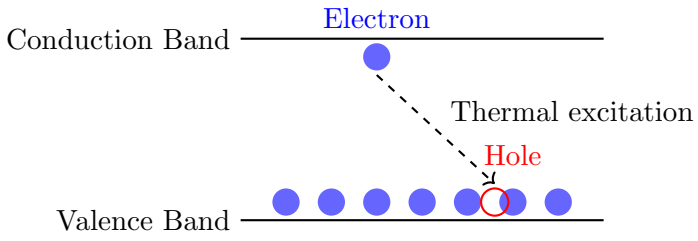
In the next sections, we will use these dispersion relations and effective masses to understand how electrons populate energy bands, how they respond to electric fields, and ultimately how they enable the operation of diodes and transistors.

5 Holes: The Missing Electrons?

One of the most elegant and practically important concepts in semiconductor physics is the **hole**. Holes are not merely an abstract convenience—they are essential for understanding current flow in semiconductors and form the basis for entire classes of devices (p-type materials, p-n junctions, bipolar transistors). In this section, we will develop the concept of holes systematically, from intuitive picture to rigorous quantum mechanical foundation.

5.1 The Intuitive Picture: A Missing Electron

Imagine a nearly full valence band, with most electron states occupied but a few left empty. This situation arises naturally in semiconductors at room temperature, where thermal energy excites some electrons from the valence band to the conduction band.



Now consider what happens when we apply an electric field **E** to this system:

1. An electron from a neighboring site can hop into the empty state (the hole).
2. This leaves behind a new empty state at the electron's original location.
3. The empty state (hole) appears to move in the direction **opposite** to electron motion.



The hole behaves as if it were a positively charged particle moving in the direction of the electric field (opposite to electron flow). This intuitive picture is powerful, but it raises a fundamental question: **Why are holes prominent in semiconductors but not in metals?**

5.2 Why Holes Matter in Semiconductors, Not in Metals

The answer lies in the band structure and the position of the Fermi level (assumed as know in the following box).

Holes in Metals vs. Semiconductors

- **In metals**: The Fermi level lies inside a band. There are plenty of empty states *just above* the Fermi level and occupied states *just below*. When an electric field is applied, electrons near the Fermi level can easily move into nearby empty states. Current is carried primarily by electrons, and the concept of "holes" is not particularly useful because there are so many empty states available everywhere.
- **In semiconductors**: At low temperatures, the valence band is completely full and the conduction band is completely empty. No current can flow. At finite temperatures, a small number of electrons are excited to the conduction band, leaving behind an equal number of empty states (holes) in the valence band. These holes are **the only available empty states** for the remaining valence electrons to move into. Consequently, hole current dominates in the valence band.

The key insight: holes become prominent when the valence band is **nearly full** and empty states are scarce. This is precisely the situation in semiconductors, where the bandgap limits the number of thermally excited electrons.

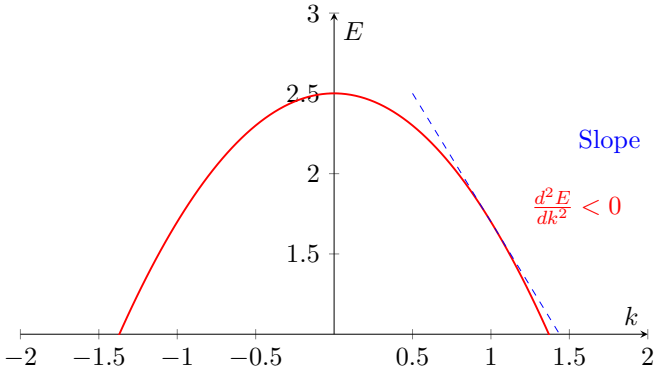
5.3 Quantum Mechanical Foundation: Negative Effective Mass

To understand holes rigorously, we must analyze the dynamics of electrons near the **top of the valence band**. Consider the dispersion relation $E(k)$ near the valence band maximum. For a simple parabolic band model:

$$E(k) = E_v - \frac{\hbar^2 k^2}{2|m^*|}$$

where E_v is the energy at the valence band top, and we explicitly write $|m^*|$ to indicate the magnitude of the effective mass. Note the **negative sign** in front of the quadratic term.

Valence Band Top - Negative Curvature



Now compute the effective mass using our definition from Section 4:

$$m^* = \frac{\hbar^2}{d^2E/dk^2}$$

Since $d^2E/dk^2 < 0$ at the valence band top, we obtain:

$$\boxed{m^* < 0}$$

This is a profound result: **Electrons near the top of the valence band have negative effective mass.**

What Does Negative Mass Mean?

Consider Newton's second law for an electron in an external electric field $\mathbf{E}$:

$$\mathbf{F} = -e\mathbf{E} = m^*\mathbf{a}$$

If m^* is negative, then the acceleration $\mathbf{a}$ is in the direction *opposite* to what we would normally expect:

$$\mathbf{a} = \frac{\mathbf{F}}{m^*} = \frac{-e\mathbf{E}}{m^*}$$

Since $m^* < 0$, the denominator is negative, making $\mathbf{a}$ positive when $\mathbf{E}$ is positive. The electron accelerates in the *same direction* as the electric field—opposite to the force on it!

This counterintuitive behavior arises entirely from the quantum mechanical $E(k)$ relation. However, dealing with negative mass particles is mathematically inconvenient. We can instead redefine our description.

5.4 The Hole as a Positive Charge Carrier

Let us transform our description by defining a new particle—the hole—with the following properties:

Definition 6 (Hole)**.** A hole is a fictitious particle that represents the absence of an electron in an otherwise full valence band. Its properties are defined as:

$$\begin{aligned} \text{Charge:} \quad & q_h = +e \\ \text{Effective mass:} \quad & m_h^* = -m_e^* > 0 \\ \text{Energy:} \quad & E_h = -E_e \quad \text{(increases downward in band diagram)} \end{aligned}$$

To see why this works, consider the dynamics. For an electron with negative mass $m_e^* < 0$:

$$\mathbf{a} = \frac{-e\mathbf{E}}{m_e^*}$$

Since m_e^* is negative, let us write $m_e^* = -m_h^*$ where $m_h^* > 0$. Then:

$$\mathbf{a} = \frac{-e\mathbf{E}}{-m_h^*} = \frac{e\mathbf{E}}{m_h^*}$$

The Hole Transformation Summary

Property	Electron at valence band top	Hole
Charge	$-e$	$+e$
Effective mass	$m_e^* < 0$	$m_h^* = -m_e^* > 0$
Velocity	$v_e = \frac{1}{\hbar}\frac{dE_e}{dk}$	$v_h = v_e$ (same)
Wave vector	$\mathbf{k}_e$	$\mathbf{k}_h = -\mathbf{k}_e$
Energy	$E_e(k)$ (max at $k = 0$)	$E_h(k) = -E_e(k)$ (min at $k = 0$)

Example 2 (Hole Motion in an Electric Field)**.** Consider an electron near the valence band top with $m_e^* = -0.5m_0$. An electric field $\mathbf{E} = 10^4$ V/m is applied in the $+x$ direction.

Electron perspective: Force on electron: $\mathbf{F} = -e\mathbf{E} = -1.6 \times 10^{-15}$ N in $+x$ direction. Acceleration: $\mathbf{a} = \mathbf{F}/m_e^* = (-1.6 \times 10^{-15})/(-0.5 \times 9.11 \times 10^{-31}) = +3.51 \times 10^{15}$ m/s^2 in $+x$ direction.

Hole perspective: A hole with $m_h^* = +0.5m_0$ experiences force $\mathbf{F} = +e\mathbf{E} = +1.6 \times 10^{-15}$ N. Acceleration: $\mathbf{a} = \mathbf{F}/m_h^* = (1.6 \times 10^{-15})/(0.5 \times 9.11 \times 10^{-31}) = +3.51 \times 10^{15}$ m/s^2 in $+x$ direction.

Both descriptions give identical physical behavior, but the hole picture uses familiar positive mass and positive charge.

5.5 Alternative Justification of the Hole Concept: Current Argument

We can also understand the hole concept by examining how current is carried in a nearly full valence band. This approach avoids the negative effective mass formulation and focuses directly on current contributions.

Current Density in Semiconductors

The current density $\mathbf{J}$ (units: A/cm^2 or A/m^2) due to a collection of charged particles is given by:

$$\mathbf{J} = q\, n\, \mathbf{v}_d$$

where:

- q is the charge per particle (coulombs)
- n is the particle concentration (cm^{-3} or m^{-3})
- $\mathbf{v}_d$ is the average drift velocity (cm/s or m/s)

In three dimensions, for a distribution of electrons with different velocities, we must sum over all electrons. The total current (units: amperes) is obtained by integrating the current density over a cross-sectional area. For our purposes, we can work with the total current contribution from a collection of electrons:

$$\mathbf{I}_{\text{total}} = -e \sum_i \mathbf{v}_i$$

where the sum runs over all electrons in the band, and $\mathbf{v}_i$ is the velocity of the i-th electron. The negative sign reflects the electron's negative charge $-e$.

Current in a Nearly Full Valence Band

Consider a valence band that is completely full except for **one missing electron** at some state. The total current carried by all electrons in this band can be written as:

$$\mathbf{I}_{\text{total}} = \mathbf{I}_{\text{full band}} - \mathbf{I}_{\text{missing electron}}$$

where:

- $\mathbf{I}_{\text{full band}} = -e \sum_{\text{all states}} \mathbf{v}_i$ is the current if the band were completely full
- $\mathbf{I}_{\text{missing electron}} = -e\, \mathbf{v}_j$ is the current that would be contributed by the missing electron if it were present (with $\mathbf{v}_j$ being its velocity)

Why a Full Band Carries No Net Current

In a perfectly periodic crystal, the electron velocity is given by the quantum mechanical **group velocity**:

$$\mathbf{v}(\mathbf{k}) = \frac{1}{\hbar} \nabla_{\mathbf{k}} E(\mathbf{k})$$

For the valence band, the dispersion relation $E(\mathbf{k})$ has a important symmetry property:

$$E(-\mathbf{k}) = E(\mathbf{k})$$

This symmetry implies that for every electron state with wavevector $\mathbf{k}$ and velocity $\mathbf{v}(\mathbf{k})$, there is a corresponding state with wavevector $-\mathbf{k}$ and velocity $\mathbf{v}(-\mathbf{k}) = -\mathbf{v}(\mathbf{k})$.

Therefore, when we sum over **all** states in a completely full valence band:

$$\mathbf{I}_{\text{full band}} = -e \sum_{\text{all } \mathbf{k}} \mathbf{v}(\mathbf{k}) = -e \left[\sum_{\mathbf{k}>0} \mathbf{v}(\mathbf{k}) + \sum_{\mathbf{k}<0} \mathbf{v}(\mathbf{k}) \right] = -e \left[\sum_{\mathbf{k}>0} \mathbf{v}(\mathbf{k}) + \sum_{\mathbf{k}>0} (-\mathbf{v}(\mathbf{k})) \right] = 0$$

Each velocity contribution is exactly canceled by its opposite counterpart. Hence, a completely full band carries **no net current**.

Current from a Missing Electron

Returning to our expression for the nearly full band:

$$\mathbf{I}_{\text{total}} = \mathbf{I}_{\text{full band}} - \mathbf{I}_{\text{missing electron}} = 0 - (-e\,\mathbf{v}_j) = +e\,\mathbf{v}_j$$

The missing electron (the empty state) thus contributes a current equivalent to a particle with:

- **Positive charge** $+e$ (instead of the electron's $-e$)
- The same velocity $\mathbf{v}_j$ as the missing electron would have had

Physical Interpretation

This derivation reveals that:

1. The current in a nearly full valence band can be completely described by considering only the **empty states** (holes).
2. Each empty state behaves as if it were a positively charged particle with charge $+e$ moving with velocity $\mathbf{v}(\mathbf{k})$.
3. We never needed to invoke negative effective mass—the hole emerges naturally from the cancellation of currents in the full band and the symmetry of the valence band $E(\mathbf{k})$ relation.
4. The hole's dynamics under external fields will follow from how $\mathbf{v}(\mathbf{k})$ changes, which is governed by the band structure.

This current-based argument provides a clean, mathematically rigorous justification for treating holes as independent positive charge carriers, complementing the negative effective mass picture developed earlier.

5.6 Experimental Consequences

The existence of holes is not merely theoretical—it has direct experimental consequences:

1. **Hall effect**: In p-type semiconductors, the Hall coefficient is positive, directly confirming that the majority carriers are positively charged.
2. **Cyclotron resonance**: Measurements of cyclotron frequency yield positive effective masses for holes, matching theoretical predictions.
3. **Thermoelectric effects**: The Seebeck coefficient changes sign between n-type and p-type materials, consistent with hole conduction.

The concept of holes thus provides a complete and consistent description of current flow in the valence band, allowing us to treat both electrons in the conduction band and holes in the valence band as ordinary positively-charged particles with positive effective masses. This duality forms the foundation for understanding p-n junctions, bipolar transistors, and virtually all semiconductor devices.

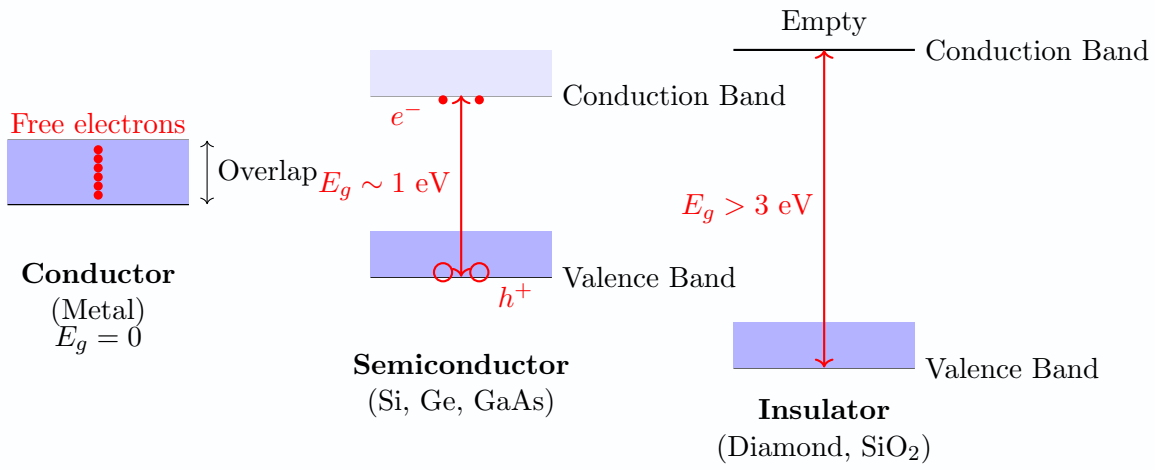
6 Classification of Solids: Insulators, Conductors, and Semiconductors

With our understanding of energy bands, effective mass, and holes, we can now systematically classify materials based on their electronic properties. This classification explains the vast differences in conductivity between materials like copper, silicon, and diamond, and it sets the stage for engineering semiconductors through doping.

6.1 Band Structure Classification

The key factor determining a material's electrical behavior is the arrangement of its energy bands—specifically, the valence band, the conduction band, and the bandgap between them.

The Three Classes of Solids



Property	Conductor	Semiconductor	Insulator
Bandgap E_g	0 (overlap)	$\sim 0.5 - 3$ eV	> 3 eV
Conductivity σ (S/m)	$10^7 - 10^8$	$10^{-6} - 10^4$	$< 10^{-10}$
Carrier concentration	$\sim 10^{22}$ cm^{-3}	$10^{10} - 10^{19}$ cm^{-3}	~ 0
Temperature dependence	σ decreases with T	σ increases with T	Insulating at all T

6.2 Detailed Analysis of Each Class

Conductors (Metals)

In conductors, either:

- The valence and conduction bands **overlap** (e.g., in alkali metals like Na, K), or
- The conduction band is **partially filled** (e.g., in noble metals like Cu, Ag, Au)

Since there is no bandgap, electrons have access to a continuum of empty states immediately above the Fermi level. Under an applied electric field, electrons can easily gain energy and move to these nearby empty states, resulting in very high conductivity.

Insulators

Insulators have a **large bandgap** ($E_g > 3$ eV). At room temperature, thermal energy $k_B T \approx 0.026$ eV is far too small to excite electrons across this gap. Consequently:

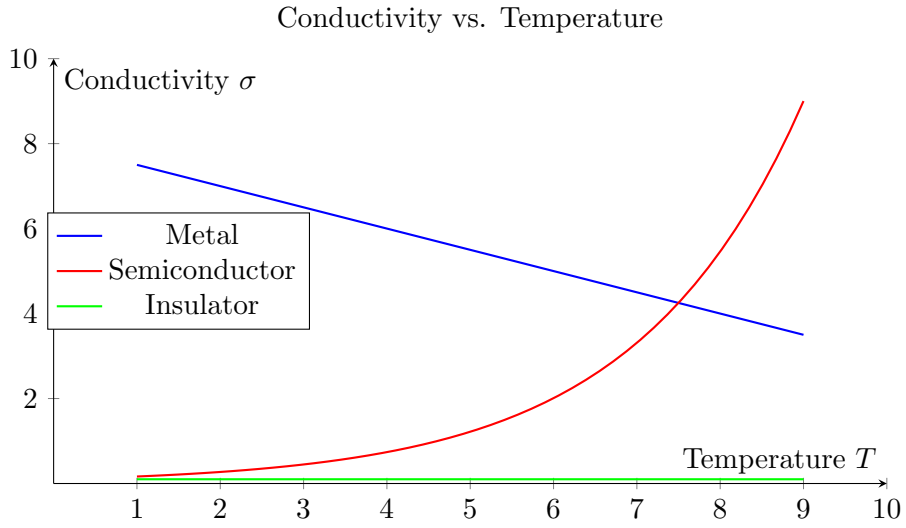
- The valence band remains completely full

- The conduction band remains completely empty
- No carriers are available for conduction

Even at elevated temperatures, the number of thermally excited electrons remains negligible. For example, diamond ($E_g = 5.5$ eV) remains an excellent insulator up to very high temperatures.

Semiconductors

Semiconductors have an **intermediate bandgap** ($\sim 0.5 - 3$ eV). At $T = 0$ K, they behave as perfect insulators—the valence band is full and the conduction band is empty. However, at finite temperatures, a significant number of electrons can be thermally excited across the gap.



6.3 Intrinsic Semiconductors

Definition 7 (Intrinsic Semiconductor)**.** An **intrinsic semiconductor** is a pure semiconductor material with no intentionally added impurities. In an intrinsic semiconductor, the number of electrons in the conduction band equals the number of holes in the valence band:

$$n = p = n_i$$

where n_i is the **intrinsic carrier concentration**.

The enormous range in n_i (seven orders of magnitude!) reflects the strong exponential dependence on bandgap. GaAs, with its larger bandgap, has far fewer intrinsic carriers than silicon, which is why it can be made semi-insulating and is useful for high-frequency devices.

Key Properties of Intrinsic Semiconductors

- **Fermi level**: Located approximately at the middle of the bandgap

$$E_F \approx \frac{E_c + E_v}{2} = E_v + \frac{E_g}{2}$$

- **Carrier concentration**: Given by the mass action law

$$n_i^2 = N_c N_v \exp\left(-\frac{E_g}{k_B T}\right)$$

  where N_c and N_v are the effective density of states in the conduction and valence bands.

- **Material examples**: Pure silicon (Si), germanium (Ge), gallium arsenide (GaAs) at room temperature

Material	n_i **at 300 K (cm^{-3})**
Germanium (Ge)	2.4×10^{13}
Silicon (Si)	1.5×10^{10}
Gallium Arsenide (GaAs)	1.8×10^{6}

6.4 Extrinsic Semiconductors and Doping

The real power of semiconductor technology comes from our ability to **engineer** the carrier concentrations by introducing controlled amounts of impurities—a process called **doping**.

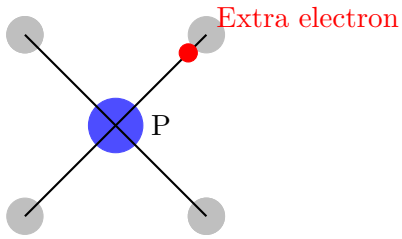
The Doping Revolution

Pure, intrinsic silicon has a carrier concentration of only $\sim 10^{10}$ cm^{-3}. By adding impurities at parts-per-million levels, we can increase this to $10^{15} - 10^{19}$ cm^{-3}—a factor of up to one billion! This ability to control conductivity over such a vast range is what makes semiconductor devices possible.

Donors and n-Type Semiconductors

Definition 8 (Donor Impurity)**.** A **donor** is an impurity atom that has more valence electrons than the host semiconductor atom it replaces. It "donates" an electron to the conduction band.

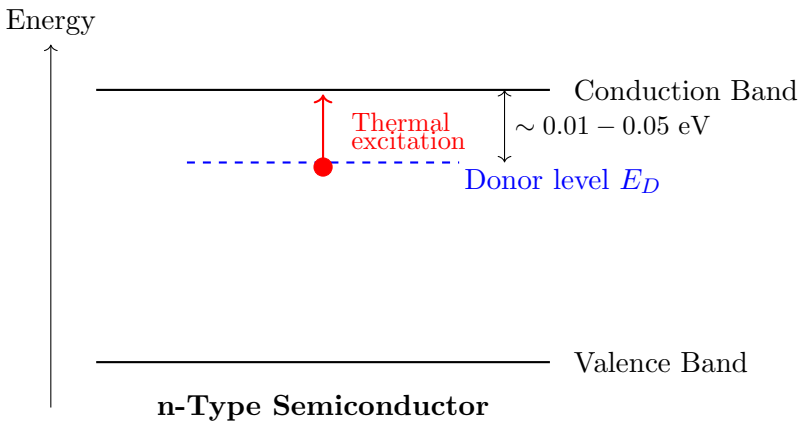
Consider silicon (Group IV, 4 valence electrons). If we replace a silicon atom with a phosphorus atom (Group V, 5 valence electrons):



Phosphorus donor in silicon

Four of phosphorus's five valence electrons form covalent bonds with neighboring silicon atoms. The fifth electron is loosely bound and can be easily excited into the conduction band at room temperature.

The energy level of this donor electron lies just below the conduction band edge—typically only $\sim 0.01 - 0.05$ eV for common donors like P, As, or Sb in Si.



Since each donor contributes one electron to the conduction band, and the donor concentration N_D is typically much larger than n_i, we have:

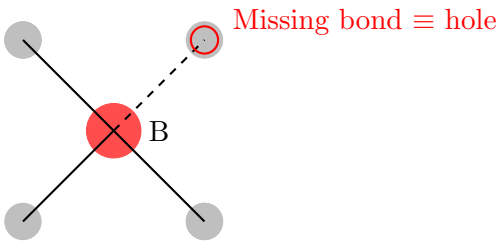
$$n \approx N_D, \quad p = \frac{n_i^2}{n} \approx \frac{n_i^2}{N_D} \ll n$$

Such a material is called **n-type** because the majority carriers are negatively charged electrons.

Acceptors and p-Type Semiconductors

Definition 9 (Acceptor Impurity)**.** An **acceptor** is an impurity atom that has fewer valence electrons than the host semiconductor atom. It "accepts" an electron from the valence band, creating a hole.

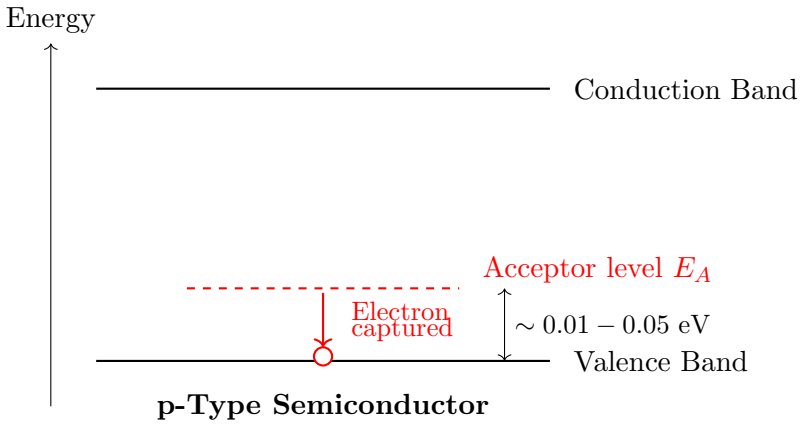
Now consider replacing a silicon atom with boron (Group III, 3 valence electrons):



Boron acceptor in silicon

Boron has only three valence electrons, so it can only form three complete covalent bonds. The fourth bond site is missing an electron—this is effectively a hole. Boron "accepts" an electron from the neighboring silicon to complete the bond, creating a hole in the valence band.

The acceptor energy level lies just above the valence band edge, typically $\sim 0.01 - 0.05$ eV for common acceptors like B, Al, or Ga in Si.



For acceptor concentration $N_A \gg n_i$, we have:

$$p \approx N_A, \quad n = \frac{n_i^2}{p} \approx \frac{n_i^2}{N_A} \ll p$$

This is **p-type** material, where the majority carriers are positively charged holes.

6.5 Compensation and Carrier Statistics

When both donors and acceptors are present, they **compensate** each other—donors provide electrons that fall into acceptor levels. The net carrier concentration is determined by the difference:

$$n \approx N_D - N_A \quad (\text{if } N_D > N_A, \text{ n-type})$$
$$p \approx N_A - N_D \quad (\text{if } N_A > N_D, \text{ p-type})$$

Doping Summary

Property	n-Type	p-Type
Dopant type	Donor	Acceptor
Dopant examples	P, As, Sb (Group V)	B, Al, Ga (Group III)
Majority carriers	Electrons ($n \approx N_D$)	Holes ($p \approx N_A$)
Minority carriers	Holes ($p = n_i^2/N_D$)	Electrons ($n = n_i^2/N_A$)
Fermi level	Near conduction band	Near valence band

- **Typical doping levels**: $10^{14} - 10^{20}$ cm^{-3}
- **Intrinsic concentration at 300K for Si**: 1.5×10^{10} cm^{-3}
- Doping can change carrier concentration by over 10 orders of magnitude!

The ability to create n-type and p-type regions in the same crystal, and to control the carrier concentrations precisely, is what enables all modern semiconductor devices—from simple diodes to complex integrated circuits. In the next sections, we will explore how these doped materials behave when brought together to form p-n junctions, the fundamental building blocks of semiconductor electronics.

7 n-Type and p-Type Semiconductors

With our understanding of donors and acceptors from the previous section, we can now define the two types of extrinsic semiconductors that form the basis of all semiconductor devices. At this stage, we will keep the discussion conceptual, focusing on what these materials are and why they matter, without delving into the statistical mechanics of carrier distributions.

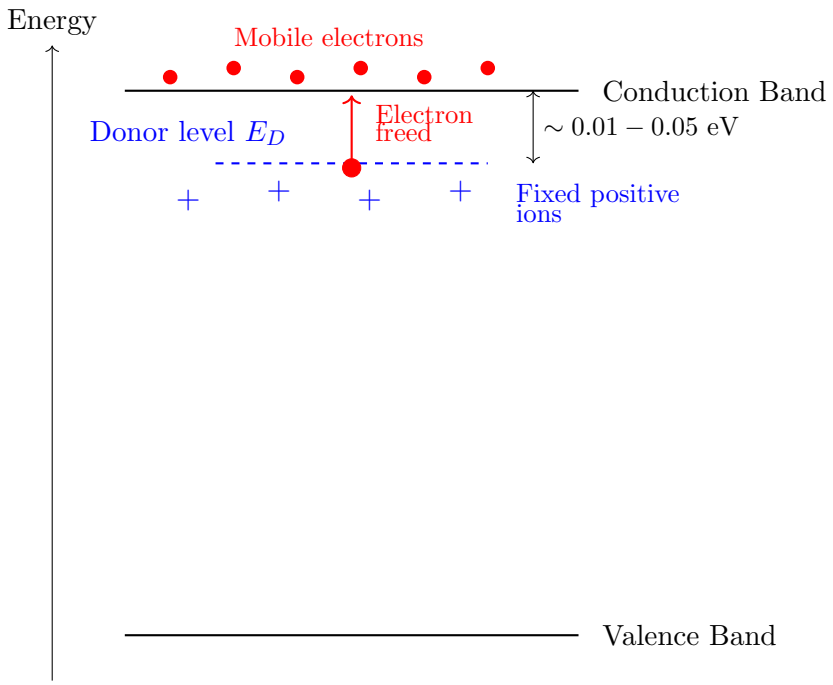
7.1 n-Type Semiconductors

Definition 10 (n-Type Semiconductor - Conceptual)**.** An **n-type semiconductor** is a semiconductor material that has been doped with **donor impurities**—atoms that have more valence electrons than the host semiconductor atoms.

The name "n-type" comes from the **negative** charge of the majority carriers (electrons).

What Makes a Material n-Type?

- **How it's made**: By adding donor impurities (Group V elements like phosphorus, arsenic, or antimony) to a pure semiconductor like silicon.
- **What happens**: Each donor atom has five valence electrons, but only four are needed to form covalent bonds with neighboring silicon atoms. The fifth electron is loosely bound and can easily break free and move around the crystal.
- **The result**: After losing its extra electron, the donor atom becomes a **positively charged ion** fixed in the crystal lattice. The freed electron becomes a mobile negative charge carrier.
- **Why "n-type"**: Because the material now has an excess of mobile **negative** charges (electrons) that can conduct electricity.



n-Type: Extra electrons, fixed positive ions

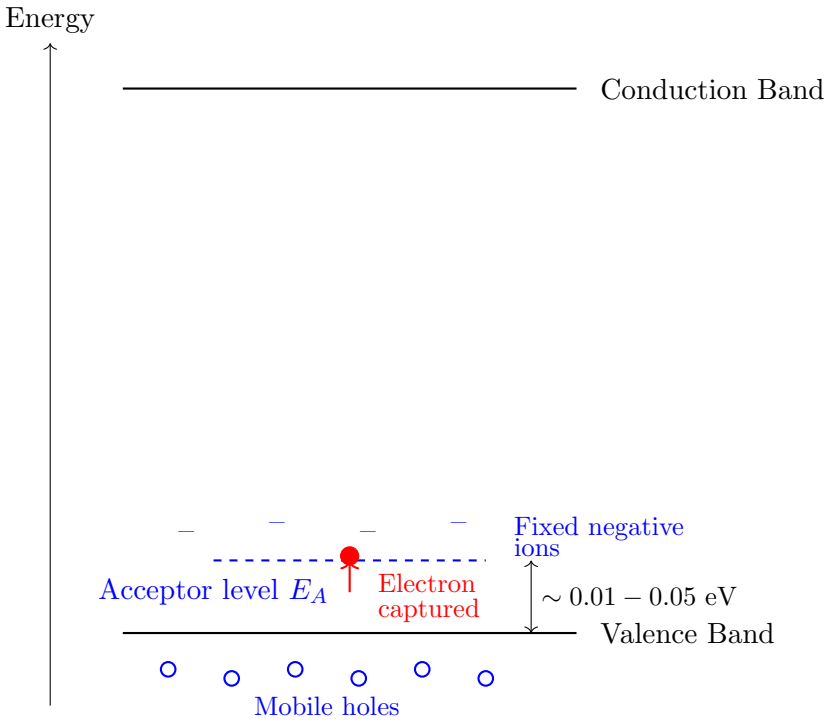
7.2 p-Type Semiconductors

Definition 11 (p-Type Semiconductor - Conceptual)**.** A **p-type semiconductor** is a semiconductor material that has been doped with **acceptor impurities**—atoms that have fewer valence electrons than the host semiconductor atoms.

The name "p-type" comes from the **positive** charge of the majority carriers (holes).

What Makes a Material p-Type?

- **How it's made**: By adding acceptor impurities (Group III elements like boron, aluminum, or gallium) to a pure semiconductor like silicon.
- **What happens**: Each acceptor atom has only three valence electrons, but needs four to complete all its covalent bonds. It "accepts" an electron from a neighboring silicon atom to complete the fourth bond.
- **The result**: After accepting an electron, the acceptor atom becomes a **negatively charged ion** fixed in the crystal lattice. The missing electron in the neighboring bond creates a **hole**—a mobile positive charge carrier.
- **Why "p-type"**: Because the material now has an excess of mobile **positive** charges (holes) that can conduct electricity.



p-Type: Extra holes, fixed negative ions

7.3 Intuitive Comparison

n-Type vs. p-Type: The Big Picture

Think of it this way:

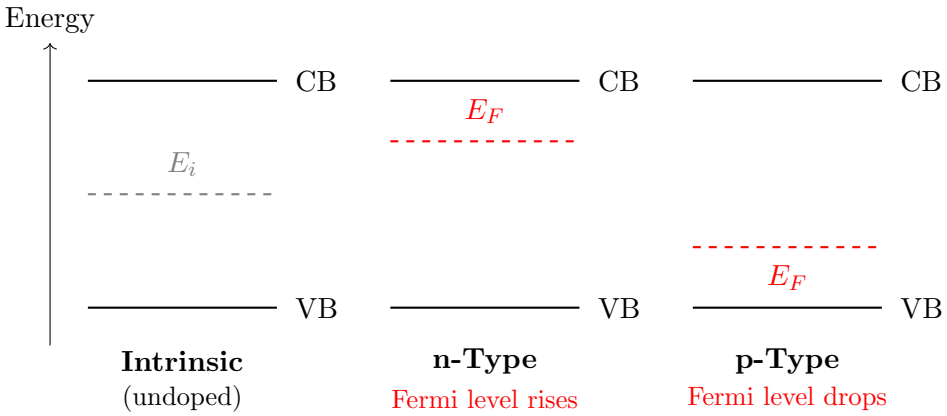
	n-Type	**p-Type**
Dopant example	Phosphorus (5 valence electrons)	Boron (3 valence electrons)
What the dopant does	"Donates" an extra electron	"Accepts" an electron, creating a hole
Mobile carriers	Electrons (negative charge)	Holes (positive charge)
Fixed ions	Positively charged donors	Negatively charged acceptors
Mnemonic	**n** for **n**egative carriers	**p** for **p**ositive carriers

- In **n-type** material, think: "Extra electrons = Negative carriers"
- In **p-type** material, think: "Missing electrons = Positive holes"

7.4 What About the Fermi Level?

Without diving into statistics, we can understand the Fermi level conceptually in doped materials:

- In **n-type** material, the extra electrons "push" the Fermi level **up** toward the conduction band. Think of it like adding more water to a reservoir—the water level rises.
- In **p-type** material, the missing electrons (holes) "pull" the Fermi level **down** toward the valence band. Think of it like removing water from a reservoir—the water level drops.
- In **intrinsic** (undoped) material, the Fermi level sits approximately in the middle of the bandgap.



Fermi Level Position in Semiconductors

7.5 Why This Matters

The creation of n-type and p-type regions in the same crystal is the fundamental building block of all semiconductor devices:

- **p-n junctions**: When n-type and p-type materials are brought together, the junction between them forms a diode—the simplest semiconductor device that allows current to flow in only one direction.

- **Transistors**: Both bipolar junction transistors (npn or pnp) and field-effect transistors rely on alternating n-type and p-type regions to provide amplification and switching.
- **Integrated circuits**: Modern chips contain billions of tiny n-type and p-type regions arranged to create complex logic circuits.

The Central Insight

The ability to create n-type and p-type materials in the same crystal, and to control where each type is located, is what enables all of modern electronics. Without this capability, we would have no diodes, no transistors, no integrated circuits—and no digital age.
In the next section, we will explore what happens when we bring n-type and p-type materials together to form a p-n junction, and discover how this simple structure can rectify current and form the basis of countless electronic devices.

8 The Fermi Energy: The "Fill Line" for Electrons

Before we can understand how electrons distribute themselves among available energy states, and how devices like p-n junctions work, we need to introduce one of the most important concepts in semiconductor physics: the **Fermi energy**.

At this point, we will avoid the mathematical complexity of Fermi-Dirac statistics. Instead, we will develop an intuitive, physical picture that captures the essence of what the Fermi energy represents and why it matters.

8.1 The Classical vs. Quantum Expectation

The Classical Expectation

To appreciate the strange behavior of electrons, let us first consider a classical system. Imagine a large classroom filled with students. Each student possesses a certain amount of thermal energy (body heat).

- **At room temperature**: The students move around with varying amounts of energy. Most will cluster around a typical average value, say E_{avg}, following a standard bell-shaped distribution.
- **As we lower the temperature**: The room gets colder. Students lose thermal energy and slow down.
- **At absolute zero ($T = 0$ K)**: In the classical world, **all students would eventually lose all their thermal energy**. They would all cluster into the absolute lowest possible energy state (imagine them all freezing perfectly still on the floor). The energy distribution would collapse into a single spike at zero.

This is how classical objects behave: given enough cold, everything settles into the absolute lowest energy state available.

The Quantum Reality: The "Strictly Assigned Seats" Rule

Now, replace the students with **electrons**. We perform the exact same experiment: we start at room temperature and gradually cool the crystal to absolute zero. What happens to their energy distribution?

The Pauli Exclusion Principle: No Sharing Allowed

Electrons are fermions, and they obey the **Pauli Exclusion Principle**. In the context of energy states, this principle acts like a strict seating chart in an auditorium:

> **Each quantum energy state can hold a maximum of two electrons**
> (one spin up, one spin down).
> No two electrons can occupy the exact same quantum state simultaneously.

This rule fundamentally changes how electrons respond to cold. They **cannot** all collapse into the lowest energy state because there is only room for two!

Because of this restriction, as we cool the system to $T = 0$ K, electrons must **stack up** in energy:

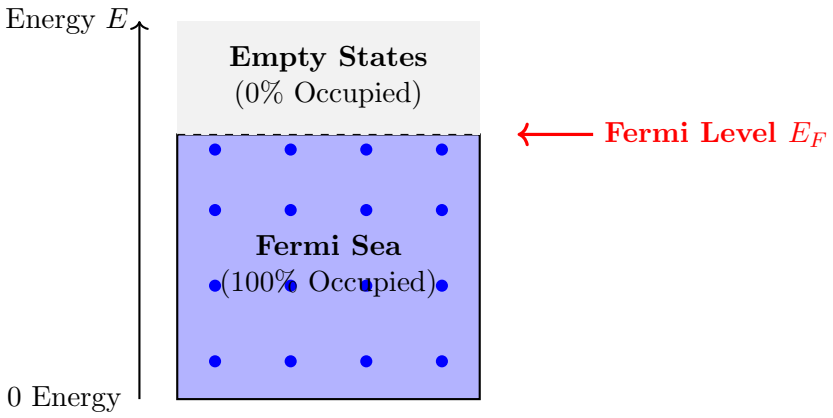
1. The lowest energy state gets filled (2 electrons).
2. The next lowest available state gets filled.
3. The next gets filled, and so on.

They continue filling state-by-state from the bottom up until every electron in the crystal has found a place.

8.2 Defining the Fermi Level

The Fermi Sea at Absolute Zero ($T = 0$ K)

This stacking creates a striking quantum phenomenon. At absolute zero, the electrons form a "Fermi sea"—a dense collection of occupied states rising from zero energy up to a maximum level, then abruptly stopping.



Electrons stack up to E_F at $T = 0$ K

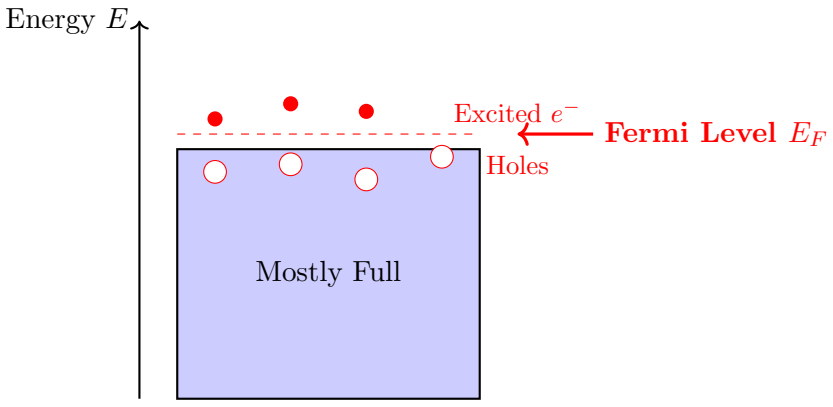
The highest energy level that contains electrons at $T = 0$ K is called the **Fermi energy** (E_F).

- **Below** E_F: Every single state is occupied.
- **Above** E_F: Every single state is empty.
- **At** E_F: The transition from "completely full" to "completely empty" is perfectly sharp.

This is the **Fermi-Dirac distribution at absolute zero**: a perfect step function. Electrons **refuse to freeze** at the bottom; they maintain a wide energy distribution up to E_F solely because they are forbidden from sharing states.

Thermal Smearing at Room Temperature ($T > 0$ K)

What happens when we warm the crystal back to room temperature? Thermal energy gives some electrons near the top of the "Fermi sea" a small kick. A tiny fraction jump above E_F, leaving behind empty spots (holes) just below it.



At $T > 0$ K, a few electrons jump above E_F

Intuitive Definition of Fermi Energy

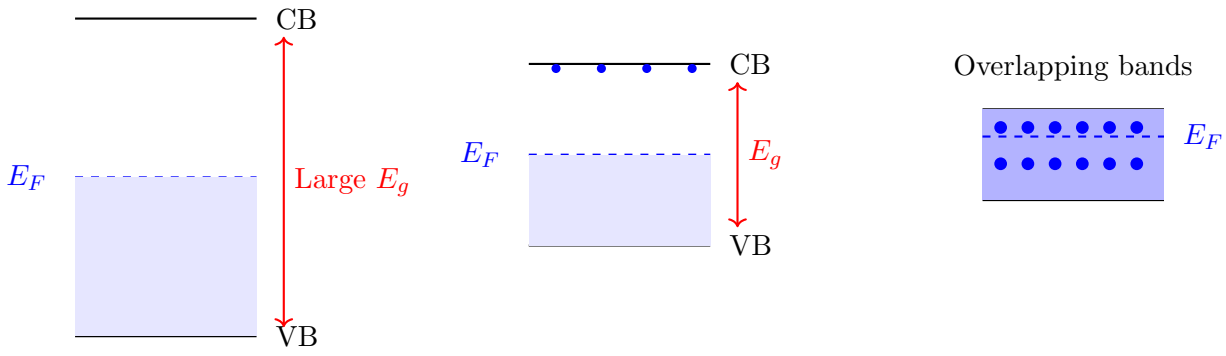
The **Fermi energy** E_F is the "fill line" or chemical potential for electrons. It acts as a rigid reference level:

- Energy states **well below** E_F are essentially **100% full**.
- Energy states **well above** E_F are essentially **100% empty**.
- The probability of finding an electron exactly at E_F is **50%** (at any $T > 0$ K).

In semiconductors, E_F typically lies inside the forbidden bandgap. Even though no actual electron states exist *at* E_F, it serves as the crucial reference point determining how many electrons are thermally excited into the conduction band.

8.3 Visualizing E_F Across Different Materials

Let us see how this "fill line" behaves in our three classes of materials:



Insulator	**Semiconductor**	**Conductor (Metal)**
E_F in gap	E_F near midgap	E_F inside band
No free electrons	Few electrons in CB	Many free electrons

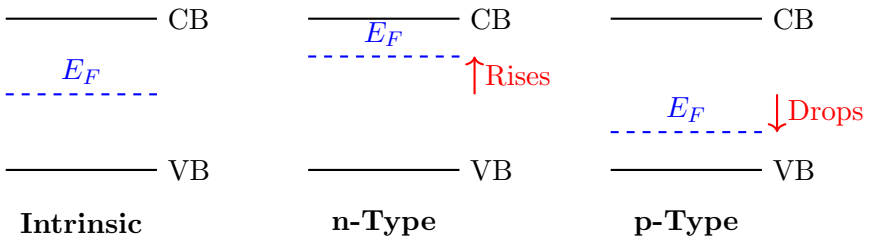
Energy increases upward ↑

8.4 What Determines E_F and Why It Matters

The position of the Fermi energy is not arbitrary; it acts as the "chemical potential" that balances the competition between quantum statistics and thermal agitation.

Factors Determining the Fermi Level

- **Doping**: Adding impurities dramatically shifts E_F. In **n-type** material (extra electrons), E_F rises toward the conduction band. In **p-type** material (extra holes), E_F drops toward the valence band.
- **Temperature**: As T increases, the sharp "fill line" softens. E_F shifts slightly to maintain charge neutrality as more carriers are thermally generated.
- **Bandgap**: Materials with larger E_g keep E_F farther from both bands, making it harder for carriers to populate the conduction/valence bands.



The Fermi energy is not just an abstract reference—it has profound practical implications:

1. **Junction formation**: When two materials with different E_F are brought together (e.g., p-type and n-type), electrons flow until their Fermi levels align. This equilibration creates the built-in electric fields essential for diodes and transistors.
2. **Carrier concentrations**: The distance between E_F and the band edges directly determines the number of free carriers:
   - E_F close to conduction band $\Rightarrow$ many electrons, few holes (n-type)
   - E_F close to valence band $\Rightarrow$ many holes, few electrons (p-type)
   - E_F far from both bands $\Rightarrow$ very few carriers (insulator/intrinsic)
3. **Device behavior**: Temperature shifts in E_F explain why semiconductor circuits behave differently in extreme environments compared to metals.

The Fermi Energy as a "Tug-of-War" Reference

Think of E_F as the rope in a tug-of-war between electrons trying to occupy low-energy states (valence band) and thermal energy trying to lift them to high-energy states (conduction band). The position of E_F tells us who is winning:

- E_F near conduction band $\Rightarrow$ electrons are winning (n-type)
- E_F near valence band $\Rightarrow$ holes are winning (p-type)
- E_F in the middle $\Rightarrow$ a draw (intrinsic)

8.5 A Simple Rule of Thumb

For most practical purposes in this course, remember:

The Fermi energy is the "fill line" for the electron sea.
States below it are generally full; states above it are generally empty.
Its distance from the band edges tells you how many carriers you have.

Example 3 (Quick Intuitive Check)**.** Consider three silicon samples at room temperature:

- Sample A: E_F is 0.05 eV below the conduction band edge.
- Sample B: E_F is exactly at midgap (0.56 eV from both bands).
- Sample C: E_F is 0.05 eV above the valence band edge.

Without calculations, we immediately identify:

- Sample A is **n-type** (fill line pushed up $\Rightarrow$ lots of electrons)
- Sample B is **intrinsic** (fill line centered $\Rightarrow$ balanced e^- and h^+)

- Sample C is **p-type** (fill line pulled down ⇒ lots of holes)

This intuitive understanding is sufficient for qualitative reasoning in device physics. The precise mathematics of the Fermi-Dirac distribution will be introduced when we need to calculate exact carrier concentrations. For now, E_F as the "maximum energy at $T = 0$ K" and the "reference fill line" will guide our analysis of junctions and devices.

9 Density of States: Counting Available Quantum States

So far, we have discussed energy bands qualitatively—we know that electrons in a semiconductor can only occupy certain energy ranges (the conduction and valence bands), and that there is a forbidden bandgap where no electron states exist. But to understand quantitatively how many electrons actually occupy these bands at a given temperature, we need to answer two questions:

1. **How many quantum states are available** at each energy? (This is the **density of states**.)
2. **What is the probability** that each available state is occupied by an electron? (This is the **Fermi-Dirac distribution**, which we will cover in the next section.)

In this section, we focus on the first question: counting the available states.

9.1 The Concept of Density of States

Definition 12 (Density of States)**.** The **density of states** $g(E)$ is a function that tells us the number of available quantum states per unit energy interval per unit volume of the crystal.

$$g(E)\,dE = \text{number of states with energy between } E \text{ and } E + dE, \text{ per unit volume}$$

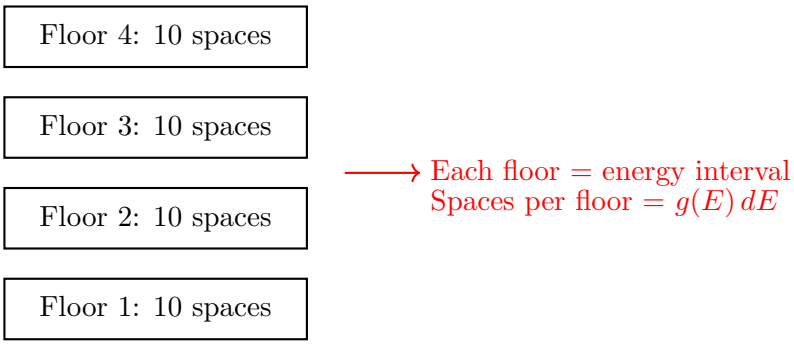
Units: $\text{cm}^{-3} \cdot \text{eV}^{-1}$ or $\text{m}^{-3} \cdot \text{J}^{-1}$.

Think of it this way: if you could look at the energy band and slice it into thin energy intervals of width dE, $g(E)\,dE$ would be the count of how many different quantum states (each capable of holding one electron, subject to the Pauli exclusion principle) exist in that slice.

9.2 Analogy: Parking Spaces in a Parking Garage

Imagine a multi-story parking garage:

- Each floor has a certain number of parking spaces.
- The "density of spaces" tells you how many parking spots are available on each floor.
- Higher floors might have the same or different numbers of spaces.
- The total number of cars that can be parked is the sum over all floors.



In our semiconductor:

- The "floors" are small energy intervals dE.
- The "parking spaces" are quantum states for electrons.
- $g(E)$ tells us how many spaces are available at each energy level.

9.3 Density of States for Free Electrons

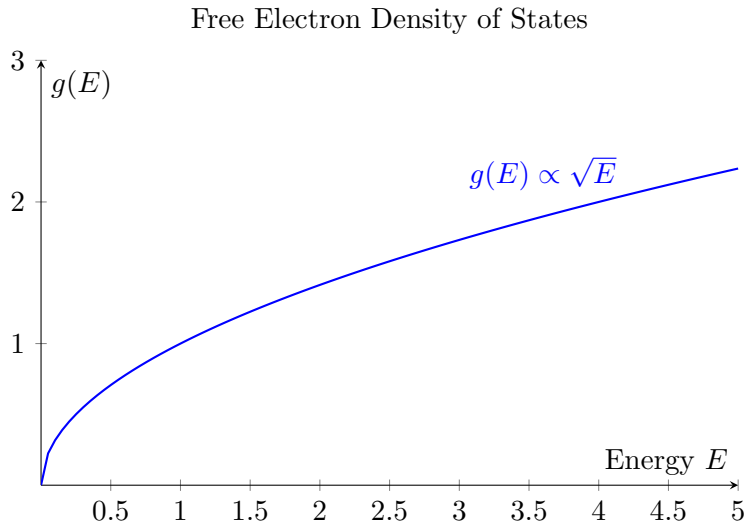
Let us begin with the simplest case: completely free electrons in a three-dimensional box. This is the situation we considered when deriving the parabolic dispersion relation $E = \hbar^2 k^2 / 2m_e$.

Density of States for Free Electrons (3D)

For free electrons in three dimensions, the density of states is:

$$g(E) = \frac{1}{2\pi^2} \left(\frac{2m_e}{\hbar^2} \right)^{3/2} \sqrt{E}$$

- This formula counts **both spin states** (spin up and spin down) for each k-state.
- It increases as $\sqrt{E}$—there are more available states at higher energies.
- The derivation comes from counting allowed k-states in k-space and converting to energy space using the dispersion relation.



9.4 Adapting the Formula for Semiconductors

In a semiconductor, electrons in the conduction band are **not completely free**—they experience the periodic potential of the crystal lattice. However, as we learned in Section 4, near the bottom of the conduction band, the dispersion relation can be approximated as parabolic:

$$E(k) = E_c + \frac{\hbar^2 k^2}{2m_e^*}$$

where E_c is the energy at the conduction band minimum, and m_e^* is the electron effective mass. This is mathematically identical to the free electron dispersion relation, except:

- The energy is shifted by E_c (the bottom of the conduction band).
- The electron mass m_e is replaced by the effective mass m_e^*.

Therefore, we can immediately write the density of states for the conduction band by making these substitutions in the free electron formula.

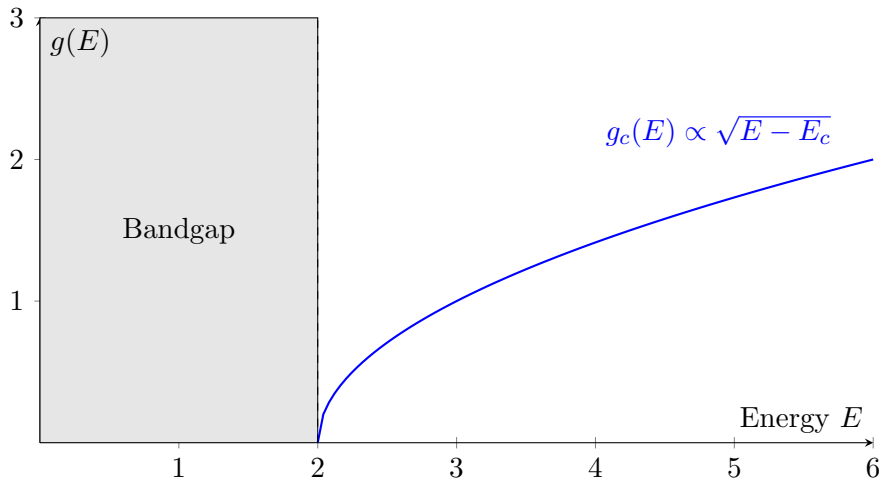
Conduction Band Density of States

$$g_c(E) = \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2}\sqrt{E - E_c} \quad \text{for } E \geq E_c$$

For energies below E_c (inside the bandgap), $g_c(E) = 0$—there are no conduction band states in the gap.
Key features:

- The density of states is **zero** below E_c (the bandgap).
- It increases as $\sqrt{E - E_c}$ above E_c.
- The effective mass m_e^* determines the curvature—a larger effective mass means a higher density of states (more states available at a given energy above the band edge).

Conduction Band Density of States



9.5 Valence Band Density of States and Holes

For the valence band, we have a similar situation but with a crucial difference. Near the top of the valence band, the dispersion relation is:

$$E(k) = E_v - \frac{\hbar^2 k^2}{2m_h^*}$$

where E_v is the energy at the valence band maximum, and m_h^* is the **hole effective mass** (positive). Notice the minus sign—energy decreases as k increases.

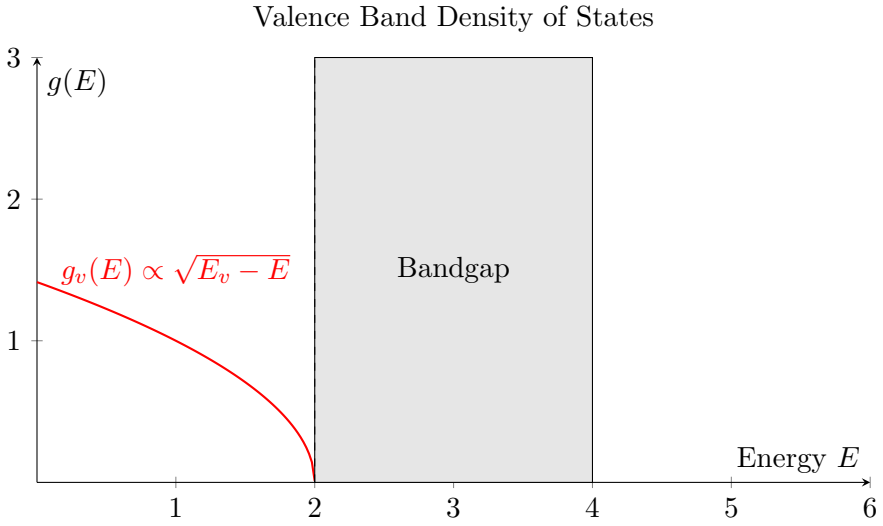
When dealing with holes, it is often more convenient to think in terms of **hole energy** increasing downward from the valence band edge. The density of states for holes in the valence band is:

Valence Band Density of States (for Holes)

$$g_v(E) = \frac{1}{2\pi^2}\left(\frac{2m_h^*}{\hbar^2}\right)^{3/2}\sqrt{E_v - E} \quad \text{for } E \leq E_v$$

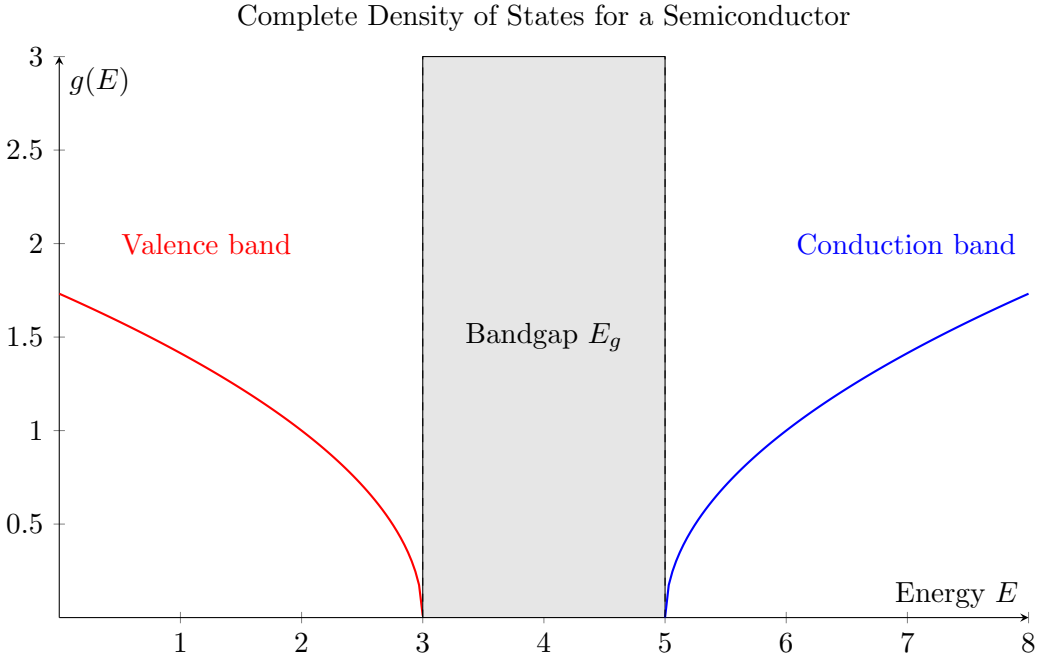
For energies above E_v, $g_v(E) = 0$.
Key features:

- The density of states increases as $\sqrt{E_v - E}$ as we go **down** into the valence band.
- The hole effective mass m_h^* plays the same role as m_e^* does for electrons.
- This formula counts states for **holes**—the absence of electrons.



9.6 Complete Picture: Conduction and Valence Bands Together

Putting both bands together gives us the complete density of states for a semiconductor:



9.7 Effective Density of States

For many practical calculations in semiconductor devices, we do not need the full energy-dependent density of states. Instead, we use a useful simplification: the **effective density of states** at the band edges.

Definition 13 (Effective Density of States)**.** The **effective density of states** N_c and N_v are defined as:

$$N_c = 2\left(\frac{2\pi m_e^* k_B T}{h^2}\right)^{3/2} \quad \text{(conduction band)}$$

$$N_v = 2\left(\frac{2\pi m_h^* k_B T}{h^2}\right)^{3/2} \quad \text{(valence band)}$$

These represent the **total number of available states** (per unit volume) within a few $k_B T$ of the band edges, treated as if they were all concentrated at the band edges.

Typical Values at Room Temperature (300 K)

Material	N_c (cm^{-3})	N_v (cm^{-3})
Silicon (Si)	2.8×10^{19}	1.04×10^{19}
Germanium (Ge)	1.04×10^{19}	6.0×10^{18}
Gallium Arsenide (GaAs)	4.7×10^{17}	7.0×10^{18}

Notice that N_c and N_v are typically on the order of 10^{19} cm^{-3}, much larger than typical doping concentrations ($10^{14} - 10^{18}$ cm^{-3}). This means that even heavily doped semiconductors have plenty of available states—the limitation on carrier concentration comes from the occupation probability, not the availability of states.

9.8 What's Next: The Occupation Probability

Now that we know **how many states are available** at each energy ($g(E)$), we need to know **how likely each state is to be occupied by an electron**. This is given by the **Fermi-Dirac distribution function** $f(E)$, which depends on:

- The energy of the state E
- The Fermi energy E_F
- The temperature T

The actual number of electrons in the conduction band is then:

$$n = \int_{E_c}^{\infty} g_c(E)\, f(E)\, dE$$

Similarly, the number of holes in the valence band is:

$$p = \int_{-\infty}^{E_v} g_v(E)\, [1 - f(E)]\, dE$$

In the next section, we will introduce the Fermi-Dirac distribution and use it together with the density of states to calculate carrier concentrations in semiconductors.

Key Takeaways

- The density of states $g(E)$ tells us how many quantum states are available at each energy.
- For the conduction band: $g_c(E) \propto \sqrt{E - E_c}$ with effective mass m_e^*.
- For the valence band (holes): $g_v(E) \propto \sqrt{E_v - E}$ with effective mass m_h^*.
- The effective densities of states N_c and N_v are convenient totals that incorporate temperature dependence.
- The actual carrier concentration depends on both the available states and the probability that those states are occupied.

10 Fermi-Dirac Distribution: The Probability of Occupation

We now have a way to count available quantum states through the density of states $g(E)$. The next question is: given these available states, what is the probability that each one is actually occupied by an electron? This is determined by quantum statistics, which governs how particles distribute themselves among available energy states in thermal equilibrium.

10.1 Quantum Statistics: Fermions and Bosons

In classical physics, particles are distinguishable and can be treated as independent entities. In quantum mechanics, however, identical particles are fundamentally **indistinguishable**. This leads to two fundamentally different classes of particles.

Definition 14 (Fermions)**. Fermions** are particles with **half-integer spin** $(\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \ldots)$. They obey the **Pauli exclusion principle**: no two fermions can occupy the same quantum state simultaneously.

Examples: electrons, protons, neutrons, quarks.

Definition 15 (Bosons)**. Bosons** are particles with **integer spin** $(0, 1, 2, \ldots)$. They do **not** obey the Pauli exclusion principle; multiple bosons can occupy the same quantum state.

Examples: photons, phonons, gluons, helium-4 atoms.

The Spin-Statistics Theorem

The connection between spin and statistics is a fundamental result of quantum field theory known as the **spin-statistics theorem** (Pauli, 1940):

Particles with half-integer spin are fermions and obey Fermi-Dirac statistics. Particles with integer spin are bosons and obey Bose-Einstein statistics.

This explains why matter is stable (electrons cannot all collapse into the lowest energy state) and why lasers work (photons are bosons and can occupy the same state).

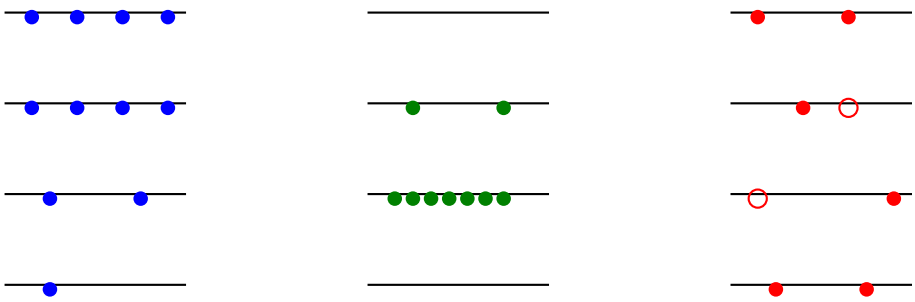
10.2 The Three Fundamental Distribution Functions

Depending on the type of particles and the conditions, three different statistical distributions describe how particles occupy energy states.

1. **Maxwell-Boltzmann distribution**: Classical statistics for distinguishable particles at low density and high temperature.

2. **Bose-Einstein distribution**: Quantum statistics for bosons (integer spin). Allows multiple particles in the same state.

3. **Fermi-Dirac distribution**: Quantum statistics for fermions (half-integer spin). Enforces the Pauli exclusion principle.



Maxwell-Boltzmann	**Bose-Einstein**	**Fermi-Dirac**
Classical, distinguishable	Bosons, integer spin	Fermions, half-integer spin
Multiple particles per state	Multiple particles per state	Pauli exclusion
	Bose-Einstein condensation	Maximum one particle per state

10.3 Mathematical Forms

Maxwell-Boltzmann Distribution

$$\boxed{f_{MB}(E) = e^{-(E-\mu)/k_BT}}$$

Valid for classical distinguishable particles or quantum particles in the low-density, high-temperature limit ($e^{-(E-\mu)/k_BT} \ll 1$). Can exceed 1 (not a true probability).

Bose-Einstein Distribution

$$\boxed{f_{BE}(E) = \frac{1}{e^{(E-\mu)/k_BT} - 1}}$$

Applies to bosons (integer spin). Can exceed 1. Diverges as $E \to \mu$ (leads to Bose-Einstein condensation). *Note: Not used in this course.*

Fermi-Dirac Distribution

$$\boxed{f_{FD}(E) = \frac{1}{e^{(E-E_F)/k_BT} + 1}}$$

Applies to fermions (half-integer spin). Always $0 \leq f_{FD} \leq 1$ (true probability). This is the distribution we use for electrons in semiconductors.

When to Use Which Distribution

Condition	Distribution	Particle Type	Example
Classical limit (low density, high T)	Maxwell-Boltzmann	Distinguishable	Ideal gas
Bosons, any energy	Bose-Einstein	Integer spin	Photons, phonons
Fermions, any energy	Fermi-Dirac	Half-integer spin	Electrons, protons
Semiconductors (non-degenerate)	MB approximation	Electrons	Most devices
Semiconductors (degenerate)	Fermi-Dirac (exact)	Electrons	Heavily doped

10.4 The Fermi-Dirac Distribution Function

Since electrons are fermions with spin 1/2, they obey Fermi-Dirac statistics. The probability that a quantum state with energy E is occupied by an electron in thermal equilibrium is:

The Fermi-Dirac Distribution

$$\boxed{f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}}$$

where:

- E is the energy of the state
- E_F is the **Fermi energy** (chemical potential)
- $k_B = 8.617 \times 10^{-5}$ eV/K (Boltzmann's constant)
- T is the absolute temperature in Kelvin

Definition 16 (Fermi Energy)**.** The **Fermi energy** E_F is the energy at which the occupation probability is exactly 1/2:

$$f(E = E_F) = \frac{1}{2}$$

States well below E_F are almost always occupied; states well above E_F are almost always empty.

10.5 Key Properties

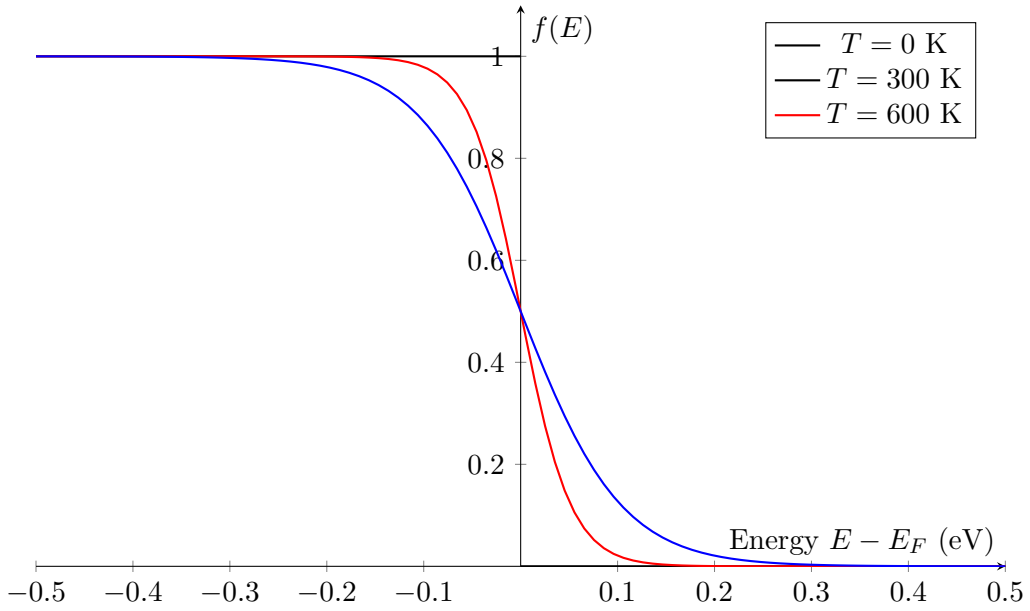
1. **Range**: $0 \leq f(E) \leq 1$ for all energies (true probability).
2. **At $T = 0$ K**: Becomes a step function:

$$f(E) = \begin{cases} 1 & \text{for } E < E_F \\ 0 & \text{for } E > E_F \end{cases}$$

3. **At $E = E_F$**: $f(E_F) = 1/2$ at any temperature.

4. **Symmetry**: $f(E_F + \Delta E) = 1 - f(E_F - \Delta E)$
5. **Temperature dependence**: The sharp step at E_F "softens" over $\sim$ a few $k_B T$.

Fermi-Dirac Distribution at Different Temperatures



10.6 Physical Interpretation

The Fermi-Dirac distribution captures two competing effects:

- **Pauli exclusion principle**: Forces electrons to occupy higher energies even at low temperatures, creating the "Fermi sea."
- **Thermal energy**: At finite temperatures, some electrons gain enough energy to be excited above E_F, leaving holes below.

Understanding the Distribution

- At $T = 0$ K: Electrons fill states up to E_F like water in a bucket.
- As T increases: The surface becomes "fuzzy" over $\sim \pm$ a few $k_B T$ around E_F.
- At room temperature: $k_B T \approx 0.026$ eV, so only electrons within ~ 0.1 eV of E_F are thermally affected.
- Electrons deep below E_F remain "frozen" by the Pauli principle.

10.7 The Maxwell-Boltzmann Approximation

For semiconductors, the important region is the **high-energy tail** where $E - E_F \gg k_B T$. In this regime:

$$f(E) = \frac{1}{1 + e^{(E-E_F)/k_B T}} \approx e^{-(E-E_F)/k_B T} \quad \text{(Maxwell-Boltzmann approximation)}$$

This applies when the Fermi level is at least a few $k_B T$ away from the band edge (non-degenerate doping).

10.8 Summary: Computing Carrier Concentrations

We now have the two essential ingredients for calculating carrier concentrations:

Density of States $g(E)$	**Fermi-Dirac Distribution** $f(E)$
How many states exist at each energy	Probability each state is occupied
Depends on effective mass and band structure	Depends on E_F and temperature
$g_c(E) \propto \sqrt{E - E_c}$	$f(E) = 1/[1 + e^{(E-E_F)/k_BT}]$

Electron concentration in the conduction band:

$$n = \int_{E_c}^{\infty} g_c(E)\, f(E)\, dE$$

Hole concentration in the valence band:

$$p = \int_{-\infty}^{E_v} g_v(E)\, [1 - f(E)]\, dE$$

In the next section, we will evaluate these integrals to obtain explicit formulas for n and p.

Key Takeaways

- Electrons are **fermions** and obey the Pauli exclusion principle.
- **Fermi-Dirac distribution** $f(E)$ gives occupation probability.
- At $T = 0$ K: step function (all states below E_F filled, above empty).
- As T increases: step "softens" over $\sim$ a few k_BT.
- For $E - E_F \gg k_BT$: reduces to exponential (Maxwell-Boltzmann approximation).
- This approximation is valid for most semiconductor conditions (non-degenerate).

11 Equilibrium Carrier Concentrations

We now have the two essential ingredients for calculating how many electrons and holes exist in a semiconductor at thermal equilibrium: the density of states $g(E)$, which tells us how many quantum states are available at each energy, and the Fermi-Dirac distribution $f(E)$, which tells us the probability that each state is occupied. In this section, we combine these to derive the fundamental expressions for electron and hole concentrations, the important $np = n_i^2$ relation, and the position of the Fermi level in intrinsic and doped semiconductors.

Notation Reference

n	Electron concentration (cm^{-3})
p	Hole concentration (cm^{-3})
N_D	Donor concentration (cm^{-3})
N_A	Acceptor concentration (cm^{-3})
N_D^+	Ionized donor concentration (cm^{-3})
N_A^-	Ionized acceptor concentration (cm^{-3})
n_i	Intrinsic carrier concentration (cm^{-3})
N_c	Effective density of states in conduction band (cm^{-3})
N_v	Effective density of states in valence band (cm^{-3})
E_c	Conduction band edge (eV)
E_v	Valence band edge (eV)
E_F	Fermi level (eV)
E_g	Bandgap energy (eV)

$$
\begin{gathered}
g(E) \times f(E) \\
\downarrow \\
n = \int g_c(E) f(E) dE \\
\downarrow \\
n = N_c e^{-(E_c - E_F)/k_B T} \\
\downarrow \\
np = n_i^2 \\
\downarrow \\
\text{Charge Neutrality: } p + N_D = n + N_A \\
\downarrow \\
\text{Exact } n, p \text{ from quadratic equation}
\end{gathered}
$$

11.1 Electron Concentration in the Conduction Band

The electron concentration n (number of electrons per unit volume) in the conduction band is obtained by integrating the product of the density of states in the conduction band and the probability that those states are occupied:

$$n = \int_{E_c}^{\infty} g_c(E)\, f(E)\, dE$$

where:

- $g_c(E)$ is the density of states in the conduction band
- $f(E)$ is the Fermi-Dirac distribution function
- E_c is the conduction band edge energy

The Density of States for the Conduction Band

From our earlier derivation, the density of states in the conduction band for a parabolic band structure is:

$$g_c(E) = \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2}\sqrt{E - E_c} \quad \text{for } E \geq E_c$$

This expression tells us that the density of states increases as the square root of the energy above the band edge.

The Fermi-Dirac Distribution in the Non-Degenerate Limit

The Fermi-Dirac distribution is:

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$$

For most semiconductors under typical operating conditions, the Fermi level E_F lies within the bandgap, at least several $k_B T$ away from both band edges. In the conduction band, $E \geq E_c$, so $E - E_F \gg k_B T$. In this case, the exponential term in the denominator dominates, and we can make the **Maxwell-Boltzmann approximation**:

$$f(E) \approx \exp\left(-\frac{E - E_F}{k_B T}\right) \quad \text{for } E - E_F \gg k_B T$$

This approximation is excellent when $E_c - E_F > 3k_B T$, which is true for non-degenerate semiconductors.

Evaluating the Integral

Substituting both expressions into our integral:

$$n = \int_{E_c}^{\infty} \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2}\sqrt{E - E_c}\,\exp\left(-\frac{E - E_F}{k_B T}\right) dE$$

We can simplify by making a change of variables. Let:

$$x = \frac{E - E_c}{k_B T} \quad \Rightarrow \quad E = E_c + x k_B T$$

Then:

$$dE = k_B T\, dx$$

$$\sqrt{E - E_c} = \sqrt{x k_B T}$$

$$\exp\left(-\frac{E - E_F}{k_B T}\right) = \exp\left(-\frac{E_c + x k_B T - E_F}{k_B T}\right) = \exp\left(-\frac{E_c - E_F}{k_B T}\right)\exp(-x)$$

Substituting everything:

$$n = \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2}\int_0^{\infty}\sqrt{x k_B T}\,\exp\left(-\frac{E_c - E_F}{k_B T}\right)\exp(-x)\,k_B T\, dx$$

Factor out the constants from the integral:

$$n = \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2}(k_B T)^{3/2}\exp\left(-\frac{E_c - E_F}{k_B T}\right)\int_0^{\infty} x^{1/2} e^{-x}\, dx$$

The integral $\int_0^{\infty} x^{1/2} e^{-x}\, dx$ is a standard form known as the Gamma function:

$$\int_0^\infty x^{1/2} e^{-x}\, dx = \Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}$$

Therefore:

$$n = \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2} (k_B T)^{3/2} \frac{\sqrt{\pi}}{2} \exp\left(-\frac{E_c - E_F}{k_B T}\right)$$

Introducing the Effective Density of States

We can simplify this expression by defining the **effective density of states in the conduction band**, N_c:

$$\boxed{N_c = 2\left(\frac{2\pi m_e^* k_B T}{h^2}\right)^{3/2}}$$

Let's verify that this matches our expression. Note that $\hbar = h/2\pi$, so:

$$\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2} = \left(\frac{2m_e^* \cdot 4\pi^2}{h^2}\right)^{3/2} = \left(\frac{8\pi^2 m_e^*}{h^2}\right)^{3/2}$$

Also, $(k_B T)^{3/2} \cdot \frac{\sqrt{\pi}}{2} \cdot \frac{1}{2\pi^2} = \frac{\sqrt{\pi}}{4\pi^2}(k_B T)^{3/2}$

Combining these carefully yields exactly $2\left(\frac{2\pi m_e^* k_B T}{h^2}\right)^{3/2}$.

Thus, the electron concentration simplifies beautifully to:

$$\boxed{n = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right)}$$

This is one of the most important equations in semiconductor physics. It tells us that the electron concentration in the conduction band depends on:

- The effective density of states N_c (which depends on effective mass and temperature)
- The distance between the conduction band edge E_c and the Fermi level E_F
- Temperature through both N_c and the exponential factor

11.2 Hole Concentration in the Valence Band

For holes, we need the probability that a state in the valence band is *empty*. Since $f(E)$ is the probability a state is occupied by an electron, the probability it is empty (i.e., occupied by a hole) is $1 - f(E)$.

The hole concentration p is:

$$p = \int_{-\infty}^{E_v} g_v(E)\,[1 - f(E)]\, dE$$

where $g_v(E)$ is the density of states in the valence band.

Density of States for the Valence Band

For the valence band, with hole effective mass m_h^*, the density of states is:

$$g_v(E) = \frac{1}{2\pi^2}\left(\frac{2m_h^*}{\hbar^2}\right)^{3/2} \sqrt{E_v - E} \quad \text{for } E \le E_v$$

The Hole Probability Function

For $E \leq E_v$, and with E_F well above the valence band edge in non-degenerate materials, $E_F - E \gg k_B T$. We can simplify $1 - f(E)$:

$$1 - f(E) = 1 - \frac{1}{1 + \exp\left(\frac{E-E_F}{k_B T}\right)} = \frac{\exp\left(\frac{E-E_F}{k_B T}\right)}{1 + \exp\left(\frac{E-E_F}{k_B T}\right)}$$

For $E_F - E \gg k_B T$, the denominator is approximately 1, so:

$$1 - f(E) \approx \exp\left(\frac{E - E_F}{k_B T}\right) = \exp\left(-\frac{E_F - E}{k_B T}\right)$$

Evaluating the Hole Integral

Substituting:

$$p = \int_{-\infty}^{E_v} \frac{1}{2\pi^2} \left(\frac{2m_h^*}{\hbar^2}\right)^{3/2} \sqrt{E_v - E}\, \exp\left(-\frac{E_F - E}{k_B T}\right) dE$$

Make a change of variables. Let:

$$x = \frac{E_v - E}{k_B T} \quad \Rightarrow \quad E = E_v - x k_B T$$

Then:

$$dE = -k_B T\, dx$$

Note that as $E \to -\infty$, $x \to \infty$, and as $E \to E_v$, $x \to 0$. The negative sign flips the integration limits:

$$p = \frac{1}{2\pi^2} \left(\frac{2m_h^*}{\hbar^2}\right)^{3/2} \int_{\infty}^{0} \sqrt{x k_B T}\, \exp\left(-\frac{E_F - (E_v - x k_B T)}{k_B T}\right) (-k_B T)\, dx$$

$$p = \frac{1}{2\pi^2} \left(\frac{2m_h^*}{\hbar^2}\right)^{3/2} \int_{0}^{\infty} \sqrt{x k_B T}\, \exp\left(-\frac{E_F - E_v}{k_B T}\right) \exp(-x)\, k_B T\, dx$$

This is exactly the same form as the electron integral, but with m_e^* replaced by m_h^* and with the exponential factor $\exp\left(-\frac{E_F - E_v}{k_B T}\right)$.

Evaluating the same Gamma function integral gives:

$$p = \frac{1}{2\pi^2} \left(\frac{2m_h^*}{\hbar^2}\right)^{3/2} (k_B T)^{3/2} \frac{\sqrt{\pi}}{2} \exp\left(-\frac{E_F - E_v}{k_B T}\right)$$

Effective Density of States for the Valence Band

Define the **effective density of states in the valence band**, N_v:

$$\boxed{N_v = 2\left(\frac{2\pi m_h^* k_B T}{h^2}\right)^{3/2}}$$

Then the hole concentration becomes:

$$\boxed{p = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)}$$

This is the companion equation to our electron concentration expression. It shows that the hole concentration depends on:

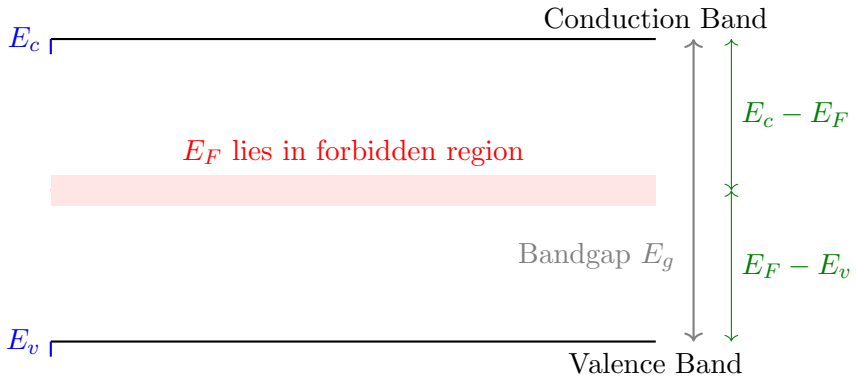
- The effective density of states N_v (determined by hole effective mass and temperature)
- The distance between the Fermi level E_F and the valence band edge E_v

11.3 Intuitive Picture: Where Do the Carriers Reside?

We now have the mathematical expressions for electron and hole concentrations. But what do these equations tell us physically? Where are the electrons and holes located in energy space? Understanding this intuitively is crucial for developing a feel for semiconductor behavior.

The Key Physical Assumption

The entire argument rests on one crucial assumption: **the Fermi level E_F lies inside the forbidden bandgap**. This is true for non-degenerate semiconductors — the materials we work with in most practical devices. Because E_F is in the bandgap, it is several k_BT away from both band edges.



The Product $g_c(E) \cdot f(E)$: Where Are the Electrons?

The number of electrons per unit energy interval at energy E in the conduction band is given by the product:

$$\text{Electron distribution} = g_c(E) \cdot f(E)$$

This product tells us how many electrons (per unit volume per unit energy) we expect to find at each energy level. Let's analyze its behavior:

1. **At $E = E_c$ (the band edge)**:

$$g_c(E_c) = 0 \quad \Rightarrow \quad \text{Product} = 0$$

   There are no states exactly at the band edge, so the electron density starts at zero.

2. **Just above E_c**:
   - $g_c(E)$ increases as $\sqrt{E - E_c}$ (opens up slowly)
   - $f(E) \approx \exp\left(-\frac{E-E_c}{k_BT}\right) \cdot \exp\left(-\frac{E_c-E_F}{k_BT}\right)$ (exponential decay)

   Initially, the square root increase dominates over the exponential decay, so the product rises from zero.

3. **Farther from E_c**:
   - Eventually, the exponential decay $\exp\left(-\frac{E-E_c}{k_BT}\right)$ becomes the dominant factor

- The product reaches a maximum and then decays toward zero

Similarly, the number of holes per unit energy interval is:

$$\text{Hole distribution} = g_v(E) \cdot [1 - f(E)]$$

For holes, the product starts at zero at $E = E_v$ (since $g_v(E_v) = 0$), rises to a maximum just below E_v, and then decays as we go deeper into the valence band.

Why This Matters: Carriers Concentrate Near Band Edges

The key insight is that **most electrons are concentrated in a narrow energy range just above** E_c, and **most holes are concentrated in a narrow energy range just below** E_v. This happens because:

- The exponential Boltzmann factor $\exp\left(-\frac{E-E_c}{k_BT}\right)$ decays very rapidly as we move away from the band edge
- At room temperature, $k_BT \approx 0.026$ eV, so the distribution decays by a factor of $e \approx 2.718$ for every 0.026 eV above E_c
- Even though $g_c(E)$ increases with energy, the exponential decay is much faster

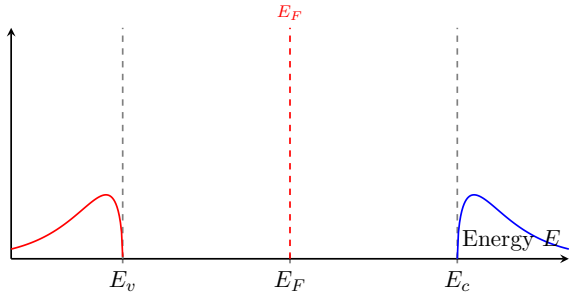
Physical Interpretation

The Fermi level acts like a "water level." Because it lies deep in the bandgap:

- Electrons in the conduction band are in the high-energy tail of the Fermi-Dirac distribution
- Their probability of occupation decreases exponentially with energy above E_c
- Consequently, almost all electrons cluster within a few k_BT of the conduction band edge
- Similarly, holes cluster within a few k_BT of the valence band edge

Visualizing the Distributions

The following figure shows how the electron and hole densities vary with energy. Note how both distributions start at zero at the band edges, rise to a peak just above/below the edges, and then decay rapidly.



The Mathematical Reason: Boltzmann Tail Dominance

Let's examine why the electron distribution peaks so close to E_c. Consider the product:

$$g_c(E) \cdot f(E) = A\sqrt{E - E_c} \cdot \exp\left(-\frac{E - E_F}{k_BT}\right)$$

where A is a constant. For $E \geq E_c$, we can write:

$$\text{Electron distribution} = A\sqrt{E-E_c}\cdot\exp\left(-\frac{E_c-E_F}{k_BT}\right)\cdot\exp\left(-\frac{E-E_c}{k_BT}\right)$$

The factor $\exp\left(-\frac{E_c-E_F}{k_BT}\right)$ is a constant that determines the overall magnitude. The energy-dependent part is:

$$h(E)=\sqrt{E-E_c}\cdot\exp\left(-\frac{E-E_c}{k_BT}\right)$$

We can find where this function peaks by taking its derivative and setting it to zero. Let $x=E-E_c\geq 0$. Then:

$$h(x)=\sqrt{x}\cdot e^{-x/k_BT}$$

Take the logarithm:

$$\ln h(x)=\frac{1}{2}\ln x-\frac{x}{k_BT}$$

Differentiate:

$$\frac{d}{dx}\ln h(x)=\frac{1}{2x}-\frac{1}{k_BT}=0 \quad\Rightarrow\quad x=\frac{k_BT}{2}$$

Thus, the peak of the electron distribution occurs at:

$$\boxed{E_{\text{peak}}=E_c+\frac{k_BT}{2}}$$

That's only half of k_BT above the band edge! At room temperature, $k_BT\approx 0.026$ eV, so the peak is a mere 0.013 eV above E_c. This is an incredibly narrow energy range.

Similarly, for holes, the peak occurs at:

$$E_{\text{peak}}=E_v-\frac{k_BT}{2}$$

just half of k_BT below the valence band edge.

Key Insight: The "Narrow Band" Approximation

Because the peak of the carrier distribution lies only $\pm k_BT/2$ from the band edges, and because the exponential tail decays extremely rapidly, we can often assume that **all carriers reside exactly at the band edges** for purposes of rough estimates and device modeling.

This justifies using the expressions $n=N_c\exp\left(-\frac{E_c-E_F}{k_BT}\right)$ and $p=N_v\exp\left(-\frac{E_F-E_v}{k_BT}\right)$ — they effectively treat all states as if they were concentrated at the band edges with an effective density N_c or N_v.

For silicon at room temperature:

- $k_BT/2\approx 0.013$ eV
- The thermal energy spread is less than 1% of the bandgap
- This is why the Boltzmann approximation works so well

11.4 The np Product and the Intrinsic Carrier Concentration

Multiplying n and p

One of the most powerful results in semiconductor physics emerges when we multiply the electron and hole concentrations:

$$np = \left[N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right)\right]\left[N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)\right]$$

Notice that the exponential factors combine:

$$\exp\left(-\frac{E_c - E_F}{k_B T}\right) \cdot \exp\left(-\frac{E_F - E_v}{k_B T}\right) = \exp\left(-\frac{E_c - E_F + E_F - E_v}{k_B T}\right) = \exp\left(-\frac{E_c - E_v}{k_B T}\right)$$

But $E_c - E_v$ is exactly the bandgap energy E_g! Therefore:

$$\boxed{np = N_c N_v \exp\left(-\frac{E_g}{k_B T}\right)}$$

The Intrinsic Carrier Concentration

In an **intrinsic** (undoped) semiconductor, every electron in the conduction band comes from a thermally excited electron that left a hole behind in the valence band. Therefore:

$$n = p = n_i$$

where n_i is called the **intrinsic carrier concentration**.
Substituting into our product relation:

$$n_i^2 = N_c N_v \exp\left(-\frac{E_g}{k_B T}\right)$$

Thus:

$$\boxed{n_i = \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2k_B T}\right)}$$

Physical Meaning of n_i

The intrinsic carrier concentration n_i is a fundamental material parameter that:

- Depends exponentially on bandgap E_g (larger bandgap $\Rightarrow$ smaller n_i)
- Increases rapidly with temperature
- Depends on effective masses through N_c and N_v
- At a given temperature, is constant for a specific material

For silicon at room temperature (300 K):

$$n_i \approx 1.5 \times 10^{10}\,\text{cm}^{-3}$$

For comparison:

- Germanium (Ge): $n_i \approx 2.4 \times 10^{13}\,\text{cm}^{-3}$
- Gallium Arsenide (GaAs): $n_i \approx 1.8 \times 10^{6}\,\text{cm}^{-3}$

The Mass Action Law

Returning to our product expression, we have the important **mass action law**:

$$\boxed{np = n_i^2}$$

This equation holds for any semiconductor in thermal equilibrium, whether intrinsic or doped. Its physical meaning is profound:

- In equilibrium, the product of electron and hole concentrations is constant at a given temperature
- If we increase n by doping, p must decrease to maintain the constant product
- Conversely, if we increase p, n must decrease

11.5 Position of the Fermi Level

We can now derive explicit expressions for the Fermi level position in intrinsic and doped semiconductors.

Fermi Level in Intrinsic Semiconductors

For an intrinsic semiconductor, $n = p = n_i$. Using our expressions:

$$N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right) = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)$$

Take natural logarithms of both sides:

$$\ln N_c - \frac{E_c - E_F}{k_B T} = \ln N_v - \frac{E_F - E_v}{k_B T}$$

Rearrange to collect terms with E_F:

$$-\frac{E_c}{k_B T} + \frac{E_F}{k_B T} + \frac{E_F}{k_B T} - \frac{E_v}{k_B T} = \ln N_v - \ln N_c$$

$$\frac{2E_F}{k_BT} - \frac{E_c + E_v}{k_BT} = \ln\left(\frac{N_v}{N_c}\right)$$

Solve for E_F:

$$\frac{2E_F}{k_BT} = \frac{E_c + E_v}{k_BT} + \ln\left(\frac{N_v}{N_c}\right)$$

$$E_F = \frac{E_c + E_v}{2} + \frac{k_BT}{2}\ln\left(\frac{N_v}{N_c}\right)$$

Recall that N_c and N_v depend on effective masses:

$$\frac{N_v}{N_c} = \left(\frac{m_h^*}{m_e^*}\right)^{3/2}$$

Therefore:

$$\boxed{E_F = \frac{E_c + E_v}{2} + \frac{3}{4}k_BT\ln\left(\frac{m_h^*}{m_e^*}\right)}$$

The first term is exactly the midpoint of the bandgap. The second term is a small correction that depends on the ratio of effective masses.

- If $m_e^* = m_h^*$, then E_F is exactly at midgap
- If $m_h^* > m_e^*$ (common in many semiconductors), the logarithm is positive, so E_F is slightly above midgap
- The correction is typically small — at room temperature, $k_BT \approx 0.026$ eV, so even a factor of 10 in mass ratio gives only about 0.045 eV shift

For most practical purposes, we can often say that the intrinsic Fermi level E_i is approximately at midgap.

Fermi Level in n-Type Semiconductors

For n-type material with donor concentration N_D, we have $n \approx N_D$. Using our expression for n:

$$N_c \exp\left(-\frac{E_c - E_F}{k_BT}\right) = N_D$$

Solve for E_F:

$$\exp\left(-\frac{E_c - E_F}{k_BT}\right) = \frac{N_D}{N_c}$$

$$-\frac{E_c - E_F}{k_BT} = \ln\left(\frac{N_D}{N_c}\right)$$

$$E_c - E_F = -k_BT\ln\left(\frac{N_D}{N_c}\right) = k_BT\ln\left(\frac{N_c}{N_D}\right)$$

Therefore:

$$\boxed{E_F = E_c - k_BT\ln\left(\frac{N_c}{N_D}\right)}$$

Since $N_D < N_c$ for non-degenerate doping, the logarithm is positive, so E_F lies below E_c. The heavier the doping, the closer E_F moves to the conduction band.

Example 4 (Fermi Level in n-Type Silicon)**.** For silicon at 300 K, $N_c \approx 2.8 \times 10^{19}\,\mathrm{cm}^{-3}$. For $N_D = 10^{16}\,\mathrm{cm}^{-3}$:

$$E_c - E_F = k_B T \ln\left(\frac{N_c}{N_D}\right) = 0.026\,\mathrm{eV} \times \ln\left(\frac{2.8 \times 10^{19}}{10^{16}}\right)$$

$$\frac{N_c}{N_D} = 2.8 \times 10^3, \quad \ln\big(2.8 \times 10^3\big) = \ln 2.8 + \ln 10^3 = 1.03 + 6.91 = 7.94$$

$$E_c - E_F = 0.026 \times 7.94 = 0.206\,\mathrm{eV}$$

So E_F is 0.206 eV below the conduction band edge. The intrinsic Fermi level E_i is approximately at midgap, which for silicon ($E_g = 1.12$ eV) is 0.56 eV below E_c. Thus doping has moved E_F significantly closer to the conduction band.

Fermi Level in p-Type Semiconductors

For p-type material with acceptor concentration N_A, we have $p \approx N_A$. Using our expression for p:

$$N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right) = N_A$$

Solve for E_F:

$$\exp\left(-\frac{E_F - E_v}{k_B T}\right) = \frac{N_A}{N_v}$$

$$-\frac{E_F - E_v}{k_B T} = \ln\left(\frac{N_A}{N_v}\right)$$

$$E_F - E_v = -k_B T \ln\left(\frac{N_A}{N_v}\right) = k_B T \ln\left(\frac{N_v}{N_A}\right)$$

Therefore:

$$\boxed{E_F = E_v + k_B T \ln\left(\frac{N_v}{N_A}\right)}$$

Here E_F lies above E_v. Heavier doping moves E_F closer to the valence band.

Example 5 (Fermi Level in p-Type Silicon)**.** For silicon at 300 K, $N_v \approx 1.04 \times 10^{19}\,\mathrm{cm}^{-3}$. For $N_A = 10^{16}\,\mathrm{cm}^{-3}$:

$$E_F - E_v = k_B T \ln\left(\frac{N_v}{N_A}\right) = 0.026\,\mathrm{eV} \times \ln\left(\frac{1.04 \times 10^{19}}{10^{16}}\right)$$

$$\frac{N_v}{N_A} = 1.04 \times 10^3, \quad \ln\big(1.04 \times 10^3\big) = \ln 1.04 + \ln 10^3 = 0.039 + 6.91 = 6.949$$

$$E_F - E_v = 0.026 \times 6.949 = 0.181\,\mathrm{eV}$$

So E_F is 0.181 eV above the valence band edge. The intrinsic Fermi level is 0.56 eV below E_c, which is equivalent to 0.56 eV above E_v (since $E_g = 1.12$ eV). Thus doping has moved E_F significantly closer to the valence band.

11.6 Charge Neutrality and Exact Carrier Concentrations

In the previous sections, we derived the equilibrium carrier concentrations using the Boltzmann approximation:

$$n = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right), \qquad p = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)$$

We also introduced the mass action law $np = n_i^2$. However, these expressions alone do not determine n and p uniquely — they depend on the unknown Fermi level E_F. To find E_F, and hence the actual carrier concentrations, we must invoke a fundamental principle: **charge neutrality**.

The Charge Neutrality Condition

In a semiconductor at equilibrium, the total charge density must be zero. Any net charge would create an electric field that would cause carrier motion until neutrality is restored. This condition is expressed as:

$$\boxed{\rho = q(p - n + N_D^+ - N_A^-) = 0}$$

where:

- q is the elementary charge (1.602×10^{-19} C)
- n is the concentration of free electrons in the conduction band
- p is the concentration of free holes in the valence band
- N_D^+ is the concentration of **ionized donors** (positively charged)
- N_A^- is the concentration of **ionized acceptors** (negatively charged)

The Complete Ionization Approximation

At room temperature, for typical doping concentrations ($N_D, N_A \lesssim 10^{18}$ cm^{-3}), the donor and acceptor energy levels are very close to the band edges. Since the Fermi level in non-degenerate semiconductors lies within the bandgap, we have E_F well below E_D in n-type material and well above E_A in p-type material. Consequently, the exponentials in the denominators are extremely small, and we can make the **complete ionization approximation**:

$$N_D^+ \approx N_D, \qquad N_A^- \approx N_A$$

This approximation holds for shallow dopants at room temperature. It fails at very low temperatures (where thermal energy is insufficient to ionize all dopants) or in degenerate semiconductors (where the Fermi level approaches the band edges). For most practical room-temperature devices, complete ionization is an excellent assumption.

The Simplified Charge Neutrality Equation

Using the complete ionization approximation, the charge neutrality condition becomes:

$$\boxed{p - n + N_D - N_A = 0}$$

or equivalently:

$$\boxed{p + N_D = n + N_A}$$

This equation states that the total positive charge (holes plus ionized donors) equals the total negative charge (electrons plus ionized acceptors).

Exact Carrier Concentrations for Doped Semiconductors

We now combine charge neutrality with the mass action law to find n and p in terms of the doping concentrations N_D and N_A.

Step 1: Express p in terms of n
From the mass action law $np = n_i^2$, we have:

$$p = \frac{n_i^2}{n}$$

Step 2: Substitute into charge neutrality
The charge neutrality equation $p + N_D = n + N_A$ becomes:

$$\frac{n_i^2}{n} + N_D = n + N_A$$

Step 3: Rearrange into a quadratic equation
Multiply both sides by n:

$$n_i^2 + N_D n = n^2 + N_A n$$

Bring all terms to one side:

$$n^2 + N_A n - N_D n - n_i^2 = 0$$

$$\boxed{n^2 + (N_A - N_D)n - n_i^2 = 0}$$

Step 4: Solve for n
The quadratic formula gives:

$$n = \frac{-(N_A - N_D) \pm \sqrt{(N_A - N_D)^2 + 4n_i^2}}{2}$$

We take the positive root (since $n > 0$):

$$\boxed{n = \frac{-(N_A - N_D) + \sqrt{(N_A - N_D)^2 + 4n_i^2}}{2}}$$

Step 5: Solve for p
Once n is known, we obtain p from the mass action law:

$$\boxed{p = \frac{n_i^2}{n}}$$

Alternatively, we could derive a similar quadratic for p by symmetry:

$$\boxed{p = \frac{-(N_D - N_A) + \sqrt{(N_D - N_A)^2 + 4n_i^2}}{2}}$$

Understanding the Square Root Expression

The expressions above look complicated, but they have a simple interpretation. Let's define the **net doping concentration**:

$$N_{\text{net}} = |N_D - N_A|$$

There are two important regimes:

Regime 1: Non-Degenerate Doping ($N_{\text{net}} \gg n_i$)

When the net doping is much larger than the intrinsic carrier concentration, the term $4n_i^2$ under the square root is negligible compared to $(N_A - N_D)^2$. In this case:

For n-type ($N_D > N_A$):

$$n \approx \frac{-(N_A - N_D) + |N_A - N_D|}{2} = \frac{(N_D - N_A) + (N_D - N_A)}{2} = N_D - N_A$$

Thus:

$$n \approx N_D - N_A, \qquad p \approx \frac{n_i^2}{N_D - N_A}$$

For p-type ($N_A > N_D$):

$$p \approx N_A - N_D, \qquad n \approx \frac{n_i^2}{N_A - N_D}$$

Regime 2: Intrinsic or Lightly Doped ($N_{\text{net}} \ll n_i$)

When the doping is very light or the temperature is high such that n_i dominates, we can expand the square root:

$$\sqrt{(N_A - N_D)^2 + 4n_i^2} \approx 2n_i\sqrt{1 + \frac{(N_A - N_D)^2}{4n_i^2}} \approx 2n_i\left(1 + \frac{(N_A - N_D)^2}{8n_i^2}\right)$$

Then:

$$n \approx \frac{-(N_A - N_D) + 2n_i + \frac{(N_A - N_D)^2}{4n_i}}{2} \approx n_i + \frac{N_D - N_A}{2}$$

For $N_D = N_A = 0$ (intrinsic), we recover $n = n_i$.

Numerical Examples

Example 1: Standard n-Type Silicon (Complete Ionization Valid)

Consider silicon at $T = 300$ K where $n_i = 1.5 \times 10^{10}$ cm^{-3}. Suppose we dope with phosphorus at $N_D = 10^{16}$ cm^{-3}, with no acceptors ($N_A = 0$).

Step 1: Check if complete ionization is valid. Since N_D is moderate and room temperature, we can assume all donors are ionized.

Step 2: Use the exact formula.

$$n = \frac{-(0 - 10^{16}) + \sqrt{(0 - 10^{16})^2 + 4(1.5 \times 10^{10})^2}}{2}$$

First compute $(N_A - N_D)^2 = (-10^{16})^2 = 10^{32}$.

Compute $4n_i^2 = 4 \times (2.25 \times 10^{20}) = 9 \times 10^{20}$.

Since $10^{32} \gg 9 \times 10^{20}$, we have:

$$\sqrt{10^{32} + 9 \times 10^{20}} \approx 10^{16}\sqrt{1 + 9 \times 10^{-12}} \approx 10^{16}(1 + 4.5 \times 10^{-12})$$

Thus:

$$n \approx \frac{10^{16} + 10^{16}(1 + 4.5 \times 10^{-12})}{2} = \frac{10^{16}(2 + 4.5 \times 10^{-12})}{2} \approx 10^{16} + 2.25 \times 10^{4}\,\text{cm}^{-3}$$

The correction term 2.25×10^4 cm^{-3} is negligible compared to 10^{16} cm^{-3}.

Step 3: Approximation yields:

$$n \approx N_D = 10^{16}\,\text{cm}^{-3}$$

$$p = \frac{n_i^2}{n} = \frac{2.25 \times 10^{20}}{10^{16}} = 2.25 \times 10^{4}\,\text{cm}^{-3}$$

Example 2: Compensated n-Type Semiconductor

Now consider a compensated semiconductor: silicon with $N_D = 10^{16}$ cm^{-3} and $N_A = 5 \times 10^{15}$ cm^{-3}.

Step 1: Compute net doping.

$$N_D - N_A = 10^{16} - 5 \times 10^{15} = 5 \times 10^{15}\,\text{cm}^{-3}$$

Step 2: Use the exact formula.

$$n = \frac{-(N_A - N_D) + \sqrt{(N_A - N_D)^2 + 4n_i^2}}{2} = \frac{5 \times 10^{15} + \sqrt{(5 \times 10^{15})^2 + 4n_i^2}}{2}$$

Compute $(N_A - N_D)^2 = (-5 \times 10^{15})^2 = 2.5 \times 10^{31}$.

$4n_i^2 = 9 \times 10^{20}$ is still negligible, so:

$$n \approx \frac{5 \times 10^{15} + 5 \times 10^{15}}{2} = 5 \times 10^{15}\,\text{cm}^{-3}$$

$$p = \frac{n_i^2}{n} = \frac{2.25 \times 10^{20}}{5 \times 10^{15}} = 4.5 \times 10^{4}\,\text{cm}^{-3}$$

Step 3: Verification of approximation. The net doping is still much larger than n_i, so the approximation $n \approx N_D - N_A$ is excellent.

Example 3: Lightly Doped n-Type (Approximation Begins to Fail)

Consider silicon at $T = 400$ K, where n_i increases significantly. For silicon at 400 K, $n_i \approx 7.7 \times 10^{12}$ cm^{-3} (rough estimate using $n_i^2 \propto T^3 e^{-E_g/k_B T}$). Take $N_D = 10^{12}$ cm^{-3} and $N_A = 0$.

Step 1: Use the exact formula.

$$n = \frac{10^{12} + \sqrt{(10^{12})^2 + 4(7.7 \times 10^{12})^2}}{2}$$

First compute $(N_A - N_D)^2 = 10^{24}$.

$4n_i^2 = 4 \times (5.93 \times 10^{25}) = 2.37 \times 10^{26}$.

Now the square root term:

$$\sqrt{10^{24} + 2.37 \times 10^{26}} = \sqrt{2.38 \times 10^{26}} = 1.543 \times 10^{13}$$

Thus:

$$n = \frac{10^{12} + 1.543 \times 10^{13}}{2} = \frac{1.643 \times 10^{13}}{2} = 8.215 \times 10^{12}\,\text{cm}^{-3}$$

Step 2: Compare with approximation $n \approx N_D$:

$$N_D = 10^{12}\,\text{cm}^{-3} \quad \text{(underestimates exact value by factor of } \sim 8.2\text{)}$$

Step 3: Compare with intrinsic value:

$$n_i = 7.7 \times 10^{12}\,\text{cm}^{-3}$$

Note that n lies between N_D and n_i, as expected. The exact value 8.22×10^{12} is slightly above n_i because the doping adds a small excess of electrons.

11.7 Practical Summary: When to Use Which Formula

Decision Tree for Carrier Concentration Calculations

1. **Identify the material and temperature** to find n_i and N_c, N_v (or use tabulated values).
2. **Check if complete ionization is valid**: For shallow dopants at room temperature with $N_D, N_A \lesssim 10^{18}$ cm^{-3}, complete ionization is an excellent approximation.
3. **Compute net doping**: $N_{\text{net}} = |N_D - N_A|$.
4. **Compare N_{net} with n_i**:
   - If $N_{\text{net}} \gg n_i$: Use the approximation

     $$n \approx N_D - N_A \quad \text{(n-type)}, \qquad p \approx \frac{n_i^2}{N_D - N_A}$$

     or

     $$p \approx N_A - N_D \quad \text{(p-type)}, \qquad n \approx \frac{n_i^2}{N_A - N_D}$$

   - If $N_{\text{net}} \lesssim n_i$: Use the exact quadratic formula:

     $$n = \frac{-(N_A - N_D) + \sqrt{(N_A - N_D)^2 + 4n_i^2}}{2}$$

   - If $N_D = N_A$ (fully compensated): The material behaves intrinsically:

     $$n = p = n_i$$

11.8 Comprehensive Summary of Key Equations

Summary: Key Equations for Equilibrium Carrier Concentrations

Effective Density of States:

$$N_c = 2\left(\frac{2\pi m_e^* k_B T}{h^2}\right)^{3/2}, \qquad N_v = 2\left(\frac{2\pi m_h^* k_B T}{h^2}\right)^{3/2}$$

Carrier Concentrations (Boltzmann Approximation):

$$n = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right), \qquad p = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)$$

Intrinsic Carrier Concentration and Mass Action Law:

$$n_i^2 = N_c N_v \exp\left(-\frac{E_g}{k_B T}\right), \qquad np = n_i^2$$

Fermi Level Positions:

$$\text{Intrinsic: } E_F = \frac{E_c + E_v}{2} + \frac{3}{4} k_B T \ln\left(\frac{m_h^*}{m_e^*}\right)$$

$$\text{n-Type: } E_F = E_c - k_B T \ln\left(\frac{N_c}{N_D}\right)$$

$$\text{p-Type: } E_F = E_v + k_B T \ln\left(\frac{N_v}{N_A}\right)$$

Charge Neutrality (Complete Ionization):

$$p + N_D = n + N_A$$

Exact Carrier Concentrations:

$$n = \frac{-(N_A - N_D) + \sqrt{(N_A - N_D)^2 + 4n_i^2}}{2}, \qquad p = \frac{n_i^2}{n}$$

Approximation for $|N_D - N_A| \gg n_i$:

$$n \approx |N_D - N_A| \quad \text{(majority carriers)}, \qquad \text{minority carriers} \approx \frac{n_i^2}{|N_D - N_A|}$$

Key Takeaways

- The product $g_c(E) \cdot f(E)$ shows that electrons concentrate within a few $k_B T$ of E_c; similarly, holes concentrate near E_v.
- The mass action law $np = n_i^2$ is universal for all semiconductors in thermal equilibrium.
- **Charge neutrality** combined with $np = n_i^2$ yields exact carrier concentrations via a quadratic equation.
- For typical room-temperature devices with moderate doping ($|N_D - N_A| \gg n_i$), the simple approximations $n \approx N_D - N_A$ (n-type) or $p \approx N_A - N_D$ (p-type) are excellent.
- The Fermi level position is determined by doping: it rises toward E_c for n-type, falls toward E_v for p-type.